TRAFFICEYE

INDIVIDUAL DRAFT THESIS

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Traffic Monitoring and Green Light Support System

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

Estimating the clearance time of a mixed-traffic queue requires vehicle composition and queue order, which a count alone does not preserve. This research develops TrafficEye as an independent traffic-monitoring and green-light-support system that connects video observations to an interpretable recommendation for a traffic officer. The final Roboflow dataset contains 3,370 images supplemented with examples representing Sri Lankan traffic. The method combines seven-class YOLOv8n detection, region-of-interest filtering, reference-line ordering, representative vehicle lengths, a published 1.177 m standstill gap, class- and position-dependent effective discharge parameters, a first-in, first-out adjustment and an additional margin. FastAPI, React and MongoDB support analysis, presentation and record storage. Live CCTV is the intended operational input; prerecorded videos demonstrate the workflow because a live installation was unavailable. The current detector checkpoint reports precision of 0.7595, recall of 0.6788, mean average precision at an intersection-over-union threshold of 0.50 of 0.7477, and mean average precision across thresholds of 0.50-0.95 of 0.5148. Recorded v2 development validation reports mAP50-95 of 0.594 for YOLOv8n and 0.537 for YOLOv5n; these results concern an earlier dataset and configuration. Analysis of 289 crossing gaps produced a 3.12-second margin. Three preliminary clearance cases from the earlier workflow yielded mean absolute error of 2.511 seconds and root mean squared error of 4.218 seconds; their footage overlaps calibration material. Scripted checks support calculation consistency, and interface captures show detection results, ordered-queue detail, recommendations and history. The contribution is an inspectable class- and position-aware vehicle-to-time workflow. The results support prototype feasibility; independent timing evaluation, officer validation and live CCTV field testing remain necessary to assess operational effectiveness.

Keywords: vehicle detection, mixed traffic, queue clearance, green-light recommendation, decision support.

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I acknowledge the researchers, dataset contributors and software developers whose publicly available work and resources supported my research and development activities. The relevant sources, datasets and software tools are credited in the methodology and references.

AI assistance used during system development and thesis preparation is disclosed in Appendix B. I remain responsible for reviewing and verifying the work presented in this thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
AP	Average Precision
ASD	Average Standstill Distance
CCTV	Closed-Circuit Television
CPU	Central Processing Unit
FIFO	First-In, First-Out
FPS	Frames Per Second
GPU	Graphics Processing Unit
HTTP	Hypertext Transfer Protocol
IoU	Intersection over Union
JSON	JavaScript Object Notation
JWT	JSON Web Token
MAE	Mean Absolute Error
mAP	Mean Average Precision
NFR	Non-functional Requirement
ORB	Oriented FAST and Rotated BRIEF
RANSAC	Random Sample Consensus
REST	Representational State Transfer
RMSE	Root Mean Squared Error
ROI	Region of Interest
SDG	Sustainable Development Goal
SPA	Single-Page Application
SUS	System Usability Scale
UAT	User Acceptance Testing
YOLO	You Only Look Once

1 INTRODUCTION

1.1 Overall Project Context

Traffic management at signalised intersections requires an understanding of the vehicles waiting to pass through a monitored approach and the time needed for that queue to clear. In mixed traffic, vehicle categories differ in their representative lengths and discharge behaviour. A vehicle count alone therefore provides limited information for estimating clearance time when vehicle composition and queue position are considered.

This thesis presents TrafficEye: Traffic Monitoring and Green Light Support System, developed as an independent research project. The project aims to use traffic-video observations to estimate queue-clearance time and recommend a green-light duration that a traffic officer can review. The research and development activities include vehicle-detection data preparation, model development, queue representation, timing calculations and a web application for presenting and storing analysis results.

TrafficEye uses a seven-class YOLOv8n detector to identify vehicles within a configured region of interest. The detected vehicles are ordered relative to a reference line. Representative vehicle lengths, class- and position-dependent effective discharge speeds, a first-in, first-out adjustment and an additional safety margin are then used to calculate the clearance estimate and green-light recommendation. Ambulance detections are also considered when producing an emergency-related timing advisory. The calculation exposes intermediate values so that the basis of a recommendation can be inspected.

Traffic signal timing involves more than identifying vehicles. The FHWA Traffic Signal Timing Manual describes the relationship between vehicle departure headways, start-up lost time and effective green time [1]. These concepts provide context for connecting traffic observations to a time estimate. TrafficEye is intended to support traffic officers through real-time traffic monitoring using live CCTV camera feeds. During the current demonstration, prerecorded traffic videos were used to show the system’s vehicle-detection, queue-analysis and timing-recommendation functions

because a live CCTV setup was not available. For future real-time deployment, live CCTV cameras can be integrated as the video source for the system's detection and timing workflow. Its intended role is to provide traffic officers with interpretable timing information to support their decisions.

1.2 Individual Component Context

TrafficEye is an independently developed web-based traffic-monitoring and green-light-support system. Its intended operational input is a live CCTV camera feed from a configured road approach. During the current demonstration, prerecorded traffic videos were uploaded to demonstrate the system's detection, queue-analysis and timing-recommendation functions because a live CCTV setup was not available.

The backend samples video frames and applies a YOLOv8n object detector to seven vehicle categories: Ambulance, Bicycle, Bus, Car, Motorcycle, Truck and Tuktuk. Detections are restricted to a configured region of interest (ROI) before being used in the queue calculation. In the current recorded-video workflow, the sampled frame containing the largest number of in-ROI detections is selected to represent the normal queue for analysis.

Vehicles in the selected frame are ordered by the perpendicular pixel distance between their bounding-box bottom-centre points and a configured reference line. Representative vehicle lengths, a standstill-gap value of 1.177 m adopted from the calibrated Average Standstill Distance reported by Gunarathne et al. [2], and class- and position-dependent lookup speeds are used to calculate clearance estimates. A first-in, first-out (FIFO) adjustment ensures that a following vehicle's calculated clearance time does not precede that of the vehicle ahead. An additional safety margin is then applied to obtain the recommended green-light duration.

The web interface presents the estimated queue length, clearance time and green-light recommendation. It also provides an ambulance-related timing advisory when an ambulance is detected within the monitored region. Intermediate calculation values allow the user to inspect the basis of the recommendation. A FastAPI backend performs the analysis, a React frontend presents the results, and MongoDB stores

analysis records. The traffic officer reviews and interprets the advisory when making a signal-timing decision.

Offline calibration and application analysis are separate workflows. Calibration provides parameters used by the timing calculation, while application analysis processes vehicle observations and presents the resulting advisory. For future real-time operation, a live CCTV feed would supply frames to the detection and timing workflow. Frame selection would need to support ongoing queue monitoring, extending the current demonstration workflow, which selects a representative frame from an uploaded recording.

1.3 Problem Statement

The research problem is how to translate vehicle observations from a fixed CCTV view into an interpretable estimate of queue-clearance time and a green-light recommendation for a traffic officer. A count describes how many vehicles are represented, but does not preserve the categories, physical composition or order of the queue. In the proposed method, those characteristics influence cumulative queue distance and the discharge parameters used to estimate time. A count alone therefore provides insufficient information for the class- and position-aware calculation investigated here.

Object-detection output also requires spatial interpretation. Categories and bounding boxes must be restricted to the monitored approach and ordered consistently. Detection errors, missed vehicles and unsuitable queue selection can propagate into timing recommendations. Evaluation therefore considers detection, queue representation and the timing calculation separately.

Ambulance-related advice introduces an additional requirement: the system must identify the vehicle category and locate the ambulance within the represented queue. The problem addressed by TrafficEye consequently joins vehicle detection, spatial queue representation, timing estimation and understandable presentation. The intended setting is live CCTV monitoring. Prerecorded footage supplies the current demonstration input, allowing the processing functions and calculations to be examined before a field installation is available.

1.4 Research Gap

Queue estimation and queue-informed signal timing are established research areas. Liu et al. estimate queues using high-resolution traffic-signal event data [3], while Ban et al. reconstruct intersection queues from mobile-sensor travel times [4]. Comert et al. examine maximum-green adjustment using queue information [5]. Lee et al. use deep-learning processes for lane-based queue estimation [6], and Umair et al. investigate video-based vehicle queue estimation [7]. These studies show that queue-state estimation has been investigated through several observation and modelling approaches.

The applied research gap investigated in TrafficEye concerns the connection between class-labelled vehicles, relative queue positions and an inspectable clearance-time advisory for an officer. The proposed workflow retains vehicle composition and ordering, constructs representative cumulative distance, selects class-position discharge parameters and exposes the intermediate calculation. Its contribution lies in developing and evaluating this combined workflow for a configured mixed-traffic stream; it does not depend on claiming that vehicle detection or queue-based timing is new.

Detector training and discharge calibration require different evidence. Labelled images support category recognition, whereas discharge calibration requires a queue observation linked to subsequent vehicle movement. The methodology separates these requirements and evaluates the perception, calculation and application layers. This separation makes it possible to examine where an error enters the video-to-time workflow and how that error affects the officer-facing recommendation.

1.5 Aim and Objectives

The aim of this independent research project is to develop and evaluate TrafficEye as an interpretable, camera-based traffic-monitoring and green-light-support system for a configured road approach. The intended use is live CCTV monitoring, with recorded-video demonstrations used to examine the system before a field installation is available.

The objectives in Table 1.1 connect the research problem to the data, detection, queue calculation, application and evaluation activities. Each objective concerns work within TrafficEye, including the normal queue advisory and ambulance-related timing support.

Table 1.1: Objectives of the delivered project

ID	Objective	Evidence sought
O1	Prepare vehicle-detection data and train/select a detector for seven classes	Versioned dataset and detector evaluation
O2	Construct an ordered queue from a configured video view	ROI, reference line and ordered detections
O3	Develop a class-position clearance calculation with a safety margin	Calibration records, equations and worked example
O4	Provide normal and ambulance-related timing information	Application response and advisory behaviour
O5	Implement a web workflow for analysis and record review	Backend, frontend and persistence functions
O6	Evaluate detection, calculation and product behaviour	Detector evaluation metrics, calculation-consistency checks, preliminary clearance cases and application tests.

1.6 Research Questions or Hypotheses

This research is guided by three research questions concerning the class- and queue-position-aware timing method, detector selection and ambulance-related timing support within TrafficEye.

RQ1: How can vehicle class and relative queue position be combined with representative vehicle geometry and calibrated effective discharge parameters to produce an interpretable queue-clearance estimate and green-light recommendation for a configured traffic stream?

RQ2: What performance trade-offs are observed between YOLOv5n and YOLOv8n when detecting the seven vehicle categories used by TrafficEye, and how do these results inform detector selection?

RQ3: How can the queue-clearance calculation be applied up to the position of a detected ambulance to provide an ambulance-related timing advisory while retaining the normal full-queue recommendation?

RQ1 addresses the principal timing method, including cumulative queue distance, class- and position-dependent effective discharge speeds, first-in, first-out consistency and an additional operational margin. RQ2 addresses the detector comparison using precision, recall, mAP50 and mAP50–95. RQ3 addresses the extension of the same timing method to the ambulance's position within the represented queue.

The study uses these research questions to organise the methodology, implementation and discussion rather than proposing formal statistical hypotheses. The detector comparison, calibration observations, calculation examples and implemented advisory workflow provide the basis for examining the questions within the demonstrated prototype scope.

1.7 Scope Limitations and Assumptions

The research covers a fixed-camera view of one configured road approach, represented as a single ordered traffic stream. It includes seven-class detection, ROI filtering, reference-line ordering, representative queue geometry, clearance estimation, normal and ambulance-related green-duration advice, and analysis history. Live CCTV is the intended operational source. Uploaded recordings demonstrate the implemented processing because a live installation was unavailable. Continuous camera ingestion and ongoing queue updates form the next deployment stage. Automatic signal actuation and multi-junction coordination are outside the current product scope.

The queue model uses representative vehicle dimensions and a common standstill gap. The 1.177 m gap is adopted from published calibration research [2]. Motorcycle, Tuktuk, Car, Bus and Truck lengths are supported by Sri Lankan vehicle-dimension measurements [8]; Bicycle and Ambulance lengths are project-defined prototype parameters. FIFO constrains the estimated clearance order. Image-space distance establishes relative queue position and is not converted into a measured road distance.

Evaluation concerns the dataset versions, camera views and recordings associated with the reported experiments. Earlier development results and the current detector checkpoint are treated separately. Preliminary clearance observations overlap calibration footage and do not establish independent accuracy of the final workflow. Transfer to different roads, lighting, weather and queue configurations requires additional evaluation.

1.8 Individual Contribution

I developed TrafficEye independently, including dataset preparation, the YOLOv5n-YOLOv8n development comparison, YOLOv8n detector integration, queue representation, discharge-calibration processing and the web application. Roboflow resources were supplemented with images representing Sri Lankan traffic. The timing method combines literature-based geometry inputs, project-derived discharge observations, an explicit FIFO adjustment and the crossing-gap-based margin.

My contribution is the integration of these elements into a vehicle-to-time workflow that presents intermediate quantities, a normal green-light recommendation and ambulance-related advice. The backend, frontend and database support analysis and record review. AI-assisted development and thesis preparation are disclosed in Appendix B; responsibility for the research, interpretation and submitted work remains with me.

1.9 Thesis Structure

Chapter 2 reviews signal timing, queue estimation, object detection, tracking and local traffic-parameter studies. Chapter 3 explains the data, research procedure, timing model and evaluation measures. Chapter 4 presents requirements and solution design, and Chapter 5 describes implementation, internal integration and testing. Chapter 6 presents the results, which are interpreted in Chapter 7. Chapter 8 considers commercialisation and sustainability. Chapter 9 concludes the work and identifies future development and evaluation. The appendices contain extended methods, proposed validation instruments and an AI-use disclosure.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This is a focused narrative review covering traffic timing, queue estimation, object detection, tracking and local traffic-parameter calibration. The sources were selected for direct relevance to the implemented workflow. Peer-reviewed publications and official technical documentation were prioritised. Publisher records were used to confirm the queue-estimation references, and the original YOLO and ByteTrack publications were consulted for their respective roles.

The review is organised around the research questions and the implemented workflow. It examines each source's traffic observations, queue representation and estimation or control output. This supports comparison between methods with different inputs while preserving the distinction between queue length, clearance time and junction-wide signal optimisation.

Table 2.1 summarises the sources and their roles in TrafficEye. Signal manuals provide operational concepts, research publications establish relevant modelling approaches, and official documentation defines the detector and dataset interfaces. Project performance is evaluated using the project results, rather than inferred from the advertised performance of a software framework.

Table 2.1: Selected literature and relevance to TrafficEye

Source	Topic	Relevance and boundary
FHWA [1]	Headways, lost time and effective green	Timing foundation; not validation of this prototype
Liu et al. [3]	Congested-intersection queue estimation	Queue-state research with a different observation method
Ban et al. [4]	Queue estimation from mobile-sensor travel times	Shows a non-video input to queue estimation
Comert et al. [5]	Queue-informed maximum-green control	Control evaluation; different scope from officer advice
Redmon et al. [9]	Unified object detection	Foundational YOLO concept, not the YOLOv8 implementation
Ultralytics [10, 11]	YOLOv8 and validation software	Detector implementation and metric definitions

Source	Topic	Relevance and boundary
Zhang et al. [12]	ByteTrack association	Offline event tracking in calibration
Gunarathne et al. [2]	Local VISSIM behaviour calibration	Published calibrated standstill-gap parameter
Jayasinghe et al. [8]	Sri Lankan vehicle dimensions	Lengths for five operational vehicle classes
Lee et al. [6]	Deep-learning lane-based queues	Spatiotemporal lane-based queue estimation
Umair et al. [7]	Video-based queue estimation	Related vision-based approach
Signal Timing Manual [13]	Operating objectives and timing	Broader signal-engineering context
Roboflow [15]	Dataset versions and export settings	Dataset preparation and reproducibility

2.2 Domain and Theoretical Foundations

2.2.1 Queue state and timing

A queue can be described by vehicle count, physical extent, order and movement state. These descriptions answer different questions. A count reports how many detected vehicles are represented. An estimate of physical extent additionally uses vehicle lengths and spacing parameters, while a clearance-time estimate connects that representation to subsequent discharge behaviour.

The FHWA manual defines start-up lost time in relation to the extra departure time of the first vehicles and discusses stabilisation of discharge headways farther back in a queue [1]. TrafficEye uses position groups to represent discharge differences, but its effective-speed method is not a direct implementation of the manual's saturation-flow procedure. The reference provides context; the project's lookup and margin require their own evidence.

2.2.2 Representative dimensions and effective speeds

Representative lengths make the calculation inspectable because the contribution of each vehicle can be shown. They also create an approximation: vehicles within one class vary in actual size. The resulting queue length should therefore be described as

an estimate based on project parameters, not as a metric measurement extracted from a bounding box.

Effective discharge speed is defined in this project as an estimated cumulative distance divided by an observed event interval. It incorporates the definition of that interval and the assumptions behind distance. It should not be interpreted as an independently measured instantaneous road speed. The choice of when the interval begins is especially important when a queue is stationary.

2.2.3 Decision support and control

Decision support supplies information that a human can review, whereas control applies a policy to the physical system. The present implementation belongs to the former category. An officer can inspect the queue and recommendation, but no demonstrated interface changes a traffic signal. This distinction determines the appropriate evaluation: correct presentation and estimation are relevant here, while junction delay or emergency-response improvement would require additional control or field studies.

The arithmetic safety margin is also distinct from an engineered signal intergreen. Amber and all-red timings are not computed in the current queue function. A recommended duration cannot therefore be treated as a complete, approved signal plan.

2.3 Related Research

2.3.1 Queue estimation and queue-informed signals

Liu et al. [3] investigate real-time queue estimation for congested signalised intersections using high-resolution event data and queue-discharge information. Ban et al. [4] estimate intersection queues from mobile-sensor travel times. Both studies demonstrate that the observation source and temporal interpretation determine how a queue is reconstructed. TrafficEye instead uses a fixed-camera vehicle representation and an explicit geometry-and-discharge calculation to produce officer-facing timing information.

Comert et al. [5] use queue-length information to adjust maximum green in an actuated-control setting evaluated through traffic simulation. Their output is a control policy, whereas TrafficEye produces a recommendation for human review. This difference matters when selecting evaluation measures: queue-clearance error and interpretation of the advisory are relevant to TrafficEye, while delay, stops and network performance require a complete control experiment. The Signal Timing Manual Second Edition also places timing decisions within operating objectives and user priorities [13].

Lee et al. [6] combine convolutional and recurrent deep-learning processes to estimate lane-based queues. Their approach models temporal traffic attributes rather than interpreting one aggregate count as a complete queue state. This supports the importance of temporal context in future live CCTV operation. TrafficEye's present timing equation remains explicit: it uses class geometry and effective discharge parameters, allowing a reader to follow the contribution of each represented vehicle.

Umair et al. [7] investigate computer-vision and deep-learning estimation of vehicle queues from urban traffic video. Their study provides a directly relevant video-based comparator and shows why a fixed view, vehicle observation and lane interpretation matter. TrafficEye's primary output extends beyond reporting queue state to presenting clearance and green-duration advice. That difference defines the applied research question; it is not a claim that video queue estimation itself is unexplored.

2.3.2 Object detection and the YOLO family

Redmon et al. [9] introduced YOLO for unified object localisation and classification. Ultralytics YOLOv8 provides the training, inference and validation interfaces used in TrafficEye [10], [11]. Lightweight YOLOv5n and YOLOv8n configurations were compared on the earlier seven-class v2 dataset. The recorded validation outputs show higher precision, mAP50 and mAP50-95 for YOLOv8n, with slightly higher recall for YOLOv5n [16]. Section 6.1.2 reports those version-specific results and distinguishes them from the later checkpoint.

Object detection supplies visual observations rather than queue-clearance times. Good validation mAP does not imply good timing accuracy, and a category error can alter

both the physical queue estimate and emergency advisory. TrafficEye therefore places detector evaluation alongside, rather than in place of, calculation and application evaluation.

2.3.3 Tracking for event evidence

ByteTrack associates detections across frames, including lower-confidence detections in its association procedure, to maintain object identities [12]. In this project, tracking is used in the offline crossing and queue-event extraction workflow. It is not part of the main uploaded-video queue-selection path.

Tracking and per-frame detection serve different purposes in this implementation. Offline tracking joins queue observations to later reference-line crossings, creating temporal records for calibration. The current application selects an ordered snapshot from sampled detections. Maintaining this distinction explains how calibration intervals are obtained without implying that the demonstrated application continuously tracks every vehicle throughout discharge. Persistent tracking is a useful extension for ongoing live CCTV queue updates.

2.3.4 Local traffic-parameter calibration

Gunarathne et al. [2] calibrate VISSIM driver-behaviour parameters for the Malabe signalised-intersection context using a genetic algorithm. Table 4 reports an Average Standstill Distance of 1.177 m for optimisation trial 43. TrafficEye adopts this published calibrated value as its standstill-gap parameter. It is a literature-based input to the physical queue calculation, rather than a distance inferred from the uploaded video's pixels.

Jayasinghe, Sivakumar and Kumarage [8] report vehicle dimensions in their study of SUMO calibration for Sri Lankan traffic. Table 1 supports the representative lengths used for Motorcycle (2.0 m), Tuktuk (2.5 m), Car (4.3 m), large Bus (10.8 m) and heavy-goods Truck (8.5 m). Bicycle (1.9 m) and Ambulance (5.5 m) are project-defined prototype values. These inputs provide a representative queue extent; they do not measure the physical dimensions of each detected vehicle.

2.4 Existing Systems, Products and Techniques

The relevant technique families include fixed or preconfigured timing, actuated signal control, sensor-based queue estimation, video analytics and simulation-based optimisation. Their roles overlap, but their outputs differ. Some provide a vehicle-presence input, some estimate queue state, and some select or actuate a complete signal policy. TrafficEye focuses on a transparent timing advisory for one represented queue.

YOLOv5n provides the detector baseline reported in the development research paper [14]. A count-based timing rule is a conceptual alternative: it uses fewer parameters but does not retain vehicle composition or queue position. TrafficEye uses these additional quantities to make the timing calculation inspectable. Comparison of timing accuracy would require the same clips, observed endpoints and observation-time definition for both methods. Commercial products are considered by function rather than ranked through an unperformed benchmark.

Table 2.2 distinguishes the evaluation question associated with each family. It is a conceptual comparison, not a benchmark result.

Table 2.2: Technique families and evaluation questions

Family	Typical role	Question relevant to TrafficEye
Detector	Vehicle category and location	Are the required vehicles correctly observed?
Tracker	Object identity over time	Can queue and crossing observations be linked?
Queue estimator	Queue count or extent	Does the representation match the visible queue?
Timing model	Clearance or phase-duration estimate	Does the estimate match an observed interval?
Signal controller	Actuation and coordination	Are delays or queues improved under a complete policy?
Officer interface	Presentation and review	Can the user understand and assess the recommendation?

2.5 Gap Synthesis and Conceptual Framework

The workflow is evaluated as a chain of dependent stages. The camera view supplies the observations, while the labelled dataset supports detector training. Detections

determine the represented queue; geometry inputs and discharge parameters determine its timing estimate; and the interface presents that estimate for officer interpretation. Figure 2.1 summarises the relationship between observations and the advisory.

The applied contribution is the combination of a seven-class visual representation and an explicit class-position calculation in an officer-facing workflow. Its transparency is useful for inspection and error analysis, but it does not remove uncertainty from the input data. This framework leads to the methodology in Chapter 3 and the separate detector, arithmetic and product results in Chapter 6.

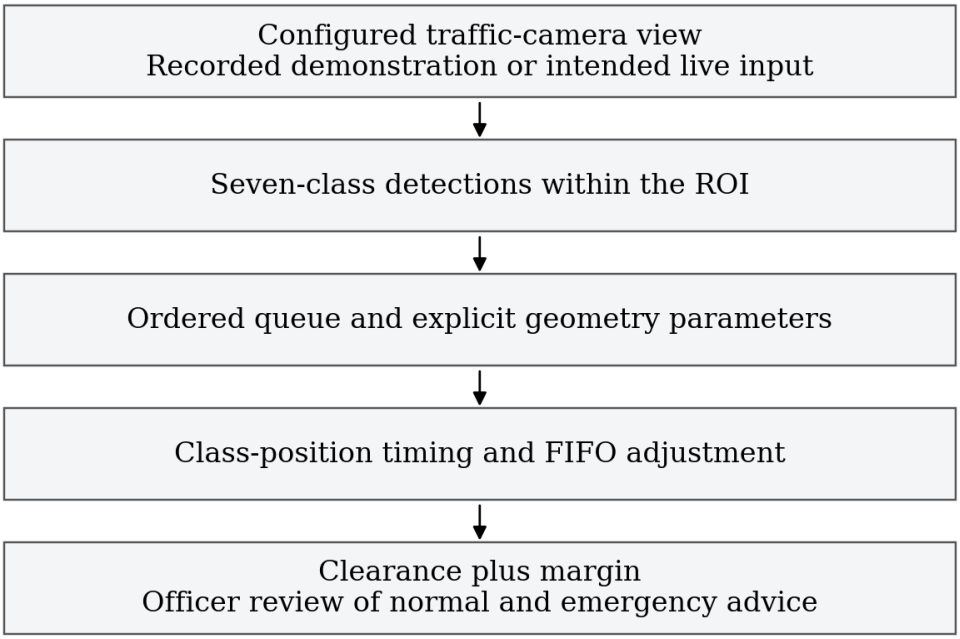


Figure 2.1: Conceptual relationship between observations and a timing advisory
Source: Author-created conceptual diagram of the TrafficEye workflow.

3 RESEARCH METHODOLOGY

3.1 Research Design

The study combines prototype development with quantitative analysis of vehicle detection and preliminary queue-clearance estimates. The artefact is a locally operated web application; the research method connects an observed vehicle queue to an inspectable timing calculation. Evaluation is divided into perception performance, arithmetic consistency and application behaviour so that success at one level is not used to imply success at another.

The methodology has two distinct data workflows. Detector development uses labelled images organised through Roboflow and exported for model training. Clearance calibration uses temporally linked queue and crossing observations from recorded clips. Figure 3.1 separates offline preparation from the operational advisory workflow.

The recorded-video demonstration provides a controlled way to inspect processing stages, reproduce an input and compare outputs while a live CCTV installation is unavailable. This choice concerns demonstration access, not a restriction of the project's intended operational purpose. Field evaluation remains a separate stage because a continuous camera stream introduces ongoing queue updates, camera availability and changing traffic conditions.

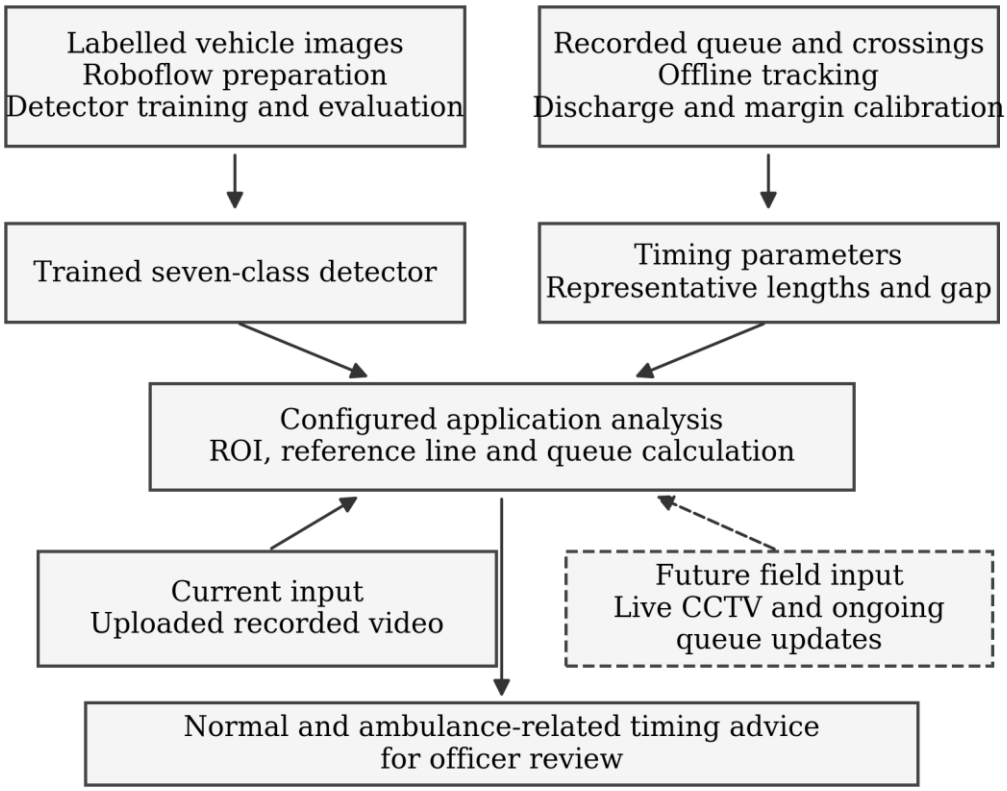


Figure 3.1: Offline preparation and operational use

Source: Author-created diagram of separate preparation workflows and application analysis. Dashed elements indicate future live CCTV integration.

3.2 Data, Participants or Study Materials

3.2.1 Final detector dataset

The final detector dataset was prepared using Roboflow data supplemented with images representing Sri Lankan traffic. This combination supports learning the seven operational categories while including local visual characteristics, such as three-wheelers and mixed vehicle composition. Images provide labelled examples for object recognition; they do not directly establish the time required for a queue to discharge.

The supplied Roboflow version record identifies Vehicle detection v3, generated on 6 October 2026, with 3,370 images: 2,377 for training, 656 for validation and 337 for testing. The checkpoint records the training path Vehicle-detection-3/data.yaml and run name yolov8n_cctv_v3. Table 3.1 gives the final image inventory, and Figure 3.2 shows the corresponding version page. The earlier 3,081-image v2 inventory is

reported separately in Section 3.2.2; current detector metrics are reported in Chapter 6.

Table 3.1: Final v3 detector dataset image inventory

Split	Images
Training	2,377
Validation	656
Test	337
Total	3,370

Source: Researcher-supplied Roboflow v3 version screenshot, generated 6 October 2026 [16].

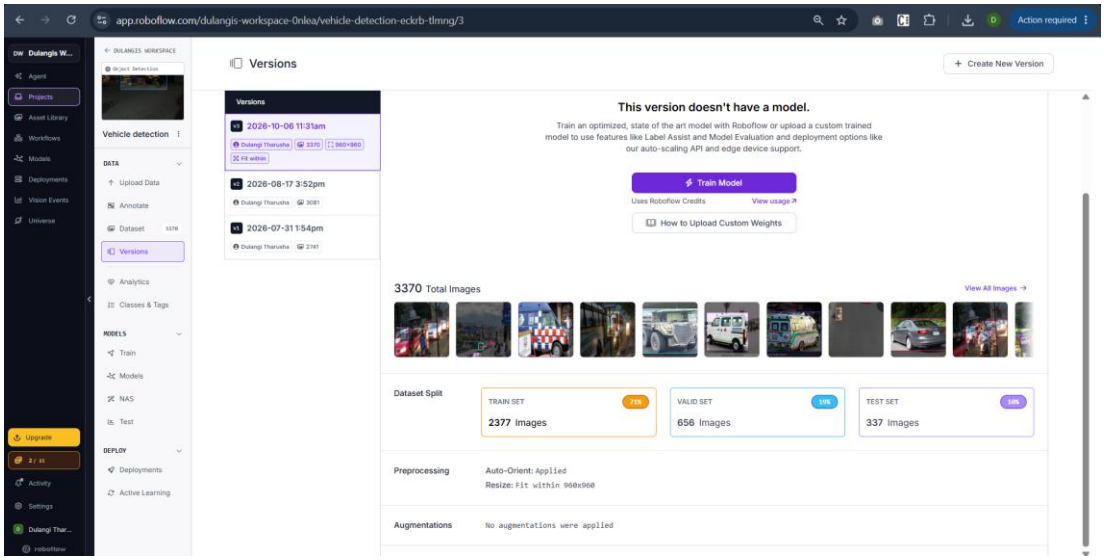


Figure 3.2: Final Roboflow v3 dataset version and preprocessing (original in colour)
Source: Author-supplied screenshot of the TrafficEye Roboflow v3 version page [16].

The version record shows auto-orientation and resizing to fit within 960 × 960 pixels. No augmentation was applied when generating this Roboflow version; training-time augmentation is a separate configuration. Roboflow versions retain preprocessing and augmentation settings with the export [15]. The checkpoint class order is Ambulance, Bicycle, Bus, Car, Motorcycle, Truck and Tuktuk. Figure 3.3 shows annotated examples with bounding boxes. Reproduction requires the exact export, class mapping and training settings together. A source-separated split should be documented for subsequent independent evaluation; it is not established by the final-version screenshot alone.

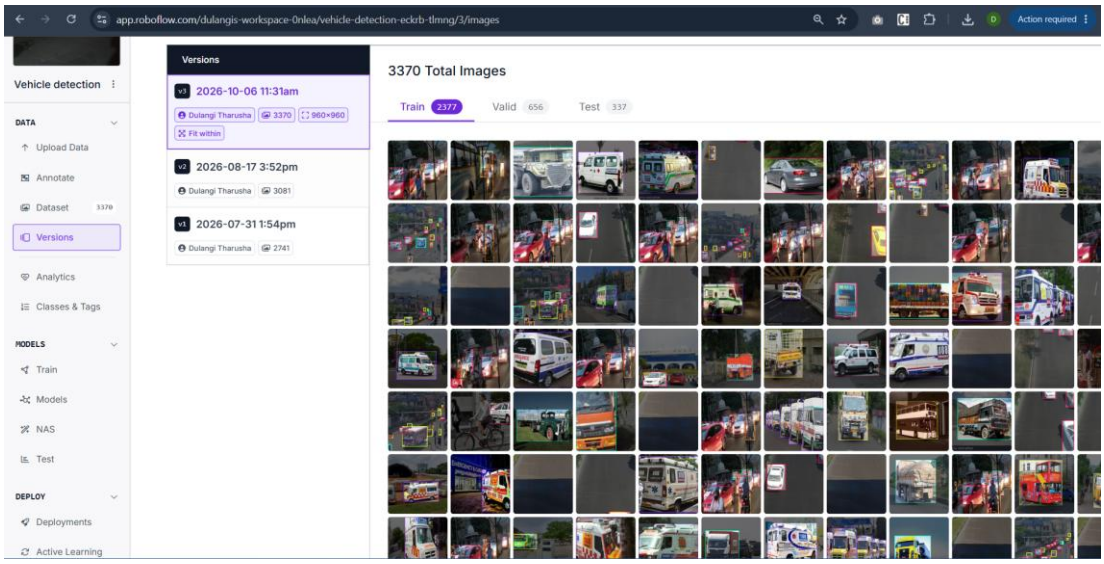


Figure 3.3: Annotated examples in the final v3 dataset (original in colour)

Source: Author-supplied screenshot of the final Roboflow v3 image gallery [16].

3.2.2 Earlier matched-comparison dataset

The development dataset contained 3,081 images: 2,174 training, 599 validation and 308 test images, as reported in the accompanying research paper [14]. The earlier export records contain 6,732 annotated instances. Table 3.2 reports the split inventory, and Table 3.3 shows the instance distribution by category. These counts describe the development dataset, not the later detector export.

The earlier split contains material from the same five source videos in training, validation and test. This source overlap creates dependence between splits even without byte-identical images. Validation results from that split can support a within-dataset comparison but provide weaker evidence of generalisation to an unseen camera or recording session.

Table 3.2: Earlier v2 dataset split inventory

Split	Images	Annotated instances
Training	2,174	4,761
Validation	599	1,337
Test	308	634
Total	3,081	6,732

Source: Development dataset split records [16] and accompanying research paper [14].

Table 3.3: Earlier v2 annotated instances by class

Class	Train	Valid	Test	Total
Ambulance	473	126	60	659
Bicycle	67	14	3	84
Bus	336	109	59	504
Car	2136	648	312	3096
Motorcycle	902	249	114	1265
Truck	473	133	61	667
Tuktuk	374	58	25	457

Source: Development dataset annotation inventory [16].

Figure 3.4 summarises the class distribution. Car contributes the largest number of annotated instances, while Bicycle has the fewest. This imbalance should be considered when interpreting aggregate detector metrics and class-specific performance.

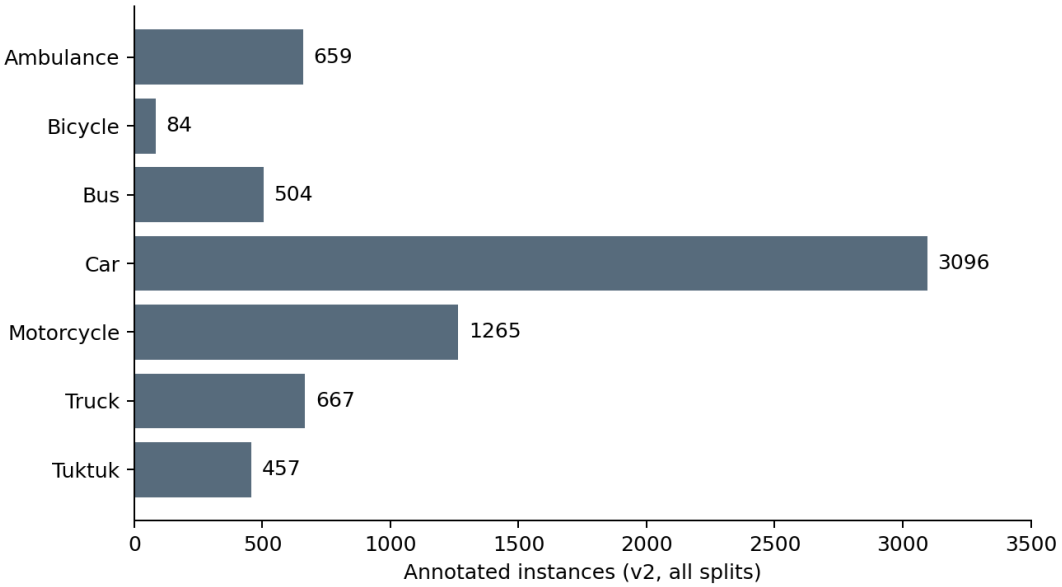


Figure 3.4: Annotated instance distribution in the earlier v2 dataset

Source: Drawn from Table 3.3 [16]. Instances are labelled vehicles, not image counts.

3.2.3 Clearance-calibration material

Discharge calibration used eleven recorded traffic clips containing 41,966 frames over approximately 1,614 seconds. The linked event records contain 344 reference-line

crossings, with 328 matched queue-to-crossing observations. Cleaning of this linked analysis set retained 283 events. Those events cover 15 of the 28 possible class-position groups and provide the descriptive calibration coverage reported in this thesis.

The analysis distinguishes observation-derived parameters from prototype values for sparsely represented combinations. Sample counts indicate how much temporal support exists for each category and position. A populated lookup entry is not, by itself, evidence of equally strong calibration across all groups. The distribution is reported in Appendix A, and incomplete coverage is considered in the discussion of timing generalisation.

The detector-image dataset and these calibration clips serve different purposes. Additional photographs may help class recognition but cannot supply vehicle crossing intervals. A calibration event requires a temporal link between the queue observation and the later reference-line crossing.

Table 3.4: Linked calibration-event inventory

Quantity	Value	Interpretation
Recorded clips	11	Temporal event material
Video frames	41,966	Total reported clip frames
Reference-line crossings	344	Crossing-log entries
Matched events	328	Crossings linked to calibration observations
Cleaned events	283	Matched events after removal of 45 rows
Supported class-position groups	15 of 28	Coverage based on source-supported cleaned events

Source: Queue-position and reference-line crossing records [16]. Counts refer to the linked analysis set.

3.2.4 Participants and stakeholder material

Traffic officers are the intended users of the timing advisory. The completed work examined recorded footage, detector outputs, timing calculations and application behaviour. Structured officer-task validation and live field observation are future evaluation stages. Appendix A provides a proposed task instrument for assessing

whether officers can interpret queue clearance, the added margin and emergency-related advice; it contains no participant results.

3.3 Experimental or Study Procedure

3.3.1 Detector preparation and model versions

The dataset labels and checkpoint class map are checked for the same seven-category order before inference. The final dataset workflow uses Roboflow resources with additional images representing Sri Lankan traffic. The detector is trained to return a class, bounding box and confidence for each object. Subsequent processing uses those outputs for ROI membership and ordering; detector training does not directly learn the clearance equation.

The current checkpoint was saved on 6 October 2026. Its recorded configuration uses yolov8n.pt initialisation, a 100-epoch training setting, image size 960, batch size 8, AdamW, patience 30 and seed 0. These are checkpoint settings. The value of 100 does not, by itself, establish the exact number of epochs completed when early stopping may be enabled.

The earlier development comparison used the seven-class v2 export, an epoch limit of 50, image size 640, batch size 16 and seed 0. Optimizer and other framework-specific defaults differed between YOLOv5n and YOLOv8n. The comparison therefore shares the dataset and main training settings without establishing identical optimisation. Current v3 metrics cannot establish improvement or deterioration against v2 because both dataset and recipe changed.

Table 3.5: Detector development and current training configuration

Property	Earlier v2 comparison	Current final checkpoint
Classes	7	7
Image size	640	960
Epoch setting	50	100
Batch size	16	8
Initial weights	YOLOv5n / YOLOv8n	yolov8n.pt

Property	Earlier v2 comparison	Current final checkpoint
Optimizer record	Framework-specific earlier settings	AdamW
Dataset inventory	Available: 3,081 images	Roboflow v3: 3,370 images; 2,377 train / 656 valid / 337 test
Role in thesis	Development configuration	Current detector version

Source: Project training configuration and checkpoint metadata [16]. Epoch values are configured limits.

3.3.2 Offline queue and crossing extraction

Offline calibration uses object tracking to link a vehicle observed in the ROI to a crossing of line_b. ByteTrack supplies track identities during this workflow [12]. Queue-position logs and crossing logs are joined through the recorded clip and track information. The interval used in the calibration is based on the queue observation or first ROI appearance and the subsequent crossing, according to the recorded pipeline.

For each event, the observed interval is the crossing timestamp minus the recorded queue-observation timestamp. The definition of that starting observation is kept consistent when deriving the corresponding effective speed. A first ROI appearance and the beginning of green are different events. The resulting quantity therefore represents effective progress from the adopted queue observation, rather than an instantaneous road speed or a directly measured saturation-flow rate.

A calibration event associates a queue observation at time t_queue with the same vehicle's reference-line crossing at time t_cross. The interval is T_obs = t_cross − t_queue. The effective discharge parameter is V_eff = D / T_obs, where D is the representative cumulative distance of that vehicle in the initial ordered queue. Valid intervals must be positive and expressed in seconds so that V_eff has units of metres per second.

Observations are grouped by vehicle category and initial position group: P1 for the first vehicle, P2 for the second, P3–4 for positions three and four, and P5+ for positions five and above. The median of available valid effective speeds provides a group summary in the calibration procedure. Each event retains its clip and track identity, connecting the observation interval to the crossing record. Sparse groups require

separate prototype treatment and further observation rather than an unsupported claim of precise calibration.

The cleaned linked observations are distributed as Car 111, Motorcycle 72, Bus 43, Tuktuk 28, Truck 27, Ambulance-labelled 2 and Bicycle 0. Class representation and queue-position coverage are uneven. The nominal ambulance events require category verification before being treated as confirmed ambulance behaviour, and Bicycle has no direct events in this analysis set. These sparse groups identify priorities for further calibration rather than supporting a claim of uniform class-level accuracy.

3.3.3 Queue geometry and length parameters

A detection is retained when its bounding-box bottom-centre lies inside the configured ROI. For the normal analysis, the backend samples every tenth frame and chooses the sampled frame with the greatest ROI vehicle count. Vehicles are ordered by ascending perpendicular pixel distance to line_b. Position 1 is closest to that line. This produces a spatially ordered snapshot rather than a tracked trajectory queue.

Table 3.6 lists representative vehicle lengths. Motorcycle, Tuktuk, Car, Bus and Truck values come from Jayasinghe et al. [8]; Bicycle and Ambulance use project-defined values [16]. The common 1.177 m gap comes from Gunarathne et al. [2]. These parameters represent queue geometry without measuring each vehicle's dimensions or spacing from pixels.

Table 3.6: Representative vehicle lengths used by the queue function

Vehicle class	Length (m)
Motorcycle	2.0
Bicycle	1.9
Tuktuk	2.5
Car	4.3
Bus	10.8
Truck	8.5
Ambulance	5.5

Source: Jayasinghe et al. [8] for five classes; prototype parameters for Bicycle and Ambulance [16], [14].

For vehicle i in a queue of n vehicles, let L_j denote the representative length of vehicle j and g denote the adopted standstill gap of 1.177 m. The cumulative distance estimate D_i is:

$$D_i = \sum_{j=1}^{i} L_j + (i - 1)g \quad (3.1)$$

D_i is expressed in metres. It represents a modelled cumulative extent and is not a direct distance-to-line measurement from the image. Queue position is mapped to P1, P2, P3-4 or P5+; the last group combines positions five and higher.

3.3.4 Clearance estimation and FIFO adjustment

The lookup supplies V_i, an effective speed in metres per second for the vehicle category and position group. Missing, unknown or non-positive entries block the calculation rather than producing an unrestricted timing output. The raw estimate is:

$$T_{raw,i} = \frac{D_i}{V_i} \quad (3.2)$$

Independently estimated group speeds can yield a shorter raw time for a vehicle farther back. The function enforces an ordered-clearance constraint:

$$T_{FIFO,1} = T_{raw,1}; \quad T_{FIFO,i} = max(T_{raw,i}, T_{FIFO,i-1}) \quad (3.3)$$

$$T_q = max_{i=1,\ldots,n} T_{FIFO,i} \quad (3.4)$$

T_q is the estimated clearance time of the represented queue. The FIFO adjustment is monotonic by construction; that property is an arithmetic guarantee, not evidence that the underlying speed estimates are accurate. Motorcycle filtering and lane changes can also depart from the single-file assumption.

3.3.5 Safety margin and recommended duration

The margin is calculated from gaps between consecutive reference-line crossings within each clip. After excluding unsuitable gaps according to the pipeline rules, 289 valid gaps remain. Their median is 2.90 s and their 85th percentile is 6.02 s. The project applies:

$$B = min(8, max(3, P_{85} - median)) = 3.12\,s \quad (3.5)$$

$$G_{rec} = round(T_q + B, 2) \quad (3.6)$$

The lower and upper bounds of 3 and 8 seconds are project design choices. The resulting 3.12-second margin is derived from the crossing-gap distribution and is added to the represented queue's clearance estimate. It is an additional operational allowance, not a substitute for traffic-engineering design of amber, all-red or pedestrian intervals. Those intervals and complete phase constraints are outside the present advisory calculation.

For the frontmost detected ambulance in an emergency candidate frame, the function uses that vehicle's FIFO-adjusted clearance estimate and adds the same margin. The normal and emergency candidate frames may differ. An emergency advisory is therefore a separate calculation, not simply a flag attached to the normal selected queue.

3.3.6 Worked example

Table 3.7 presents a development-stage example with an ordered queue of Car, Motorcycle, Motorcycle and Motorcycle. Each row shows how the class length and inter-vehicle gap contribute to cumulative distance before the effective speed is applied. This worked example explains the implemented calculation using the recorded lookup precision.

Table 3.7: Recorded clip_02 queue calculation

Pos.	Class	D (m)	V (m/s)	Raw/FIFO (s)
1	Car	4.300	2.154	1.9963
2	Motorcycle	7.477	2.389	3.1298
3	Motorcycle	10.654	3.134	3.3995
4	Motorcycle	13.831	3.134	4.4132

Source: Recorded TrafficEye calculation output [16]. Displayed speeds are rounded; times use stored precision.

The estimated clearance is 4.4132 s and the recommendation is 7.53 s after adding 3.12 s and rounding to two decimal places. The recorded observed clearance is 4.500 s, yielding an error of approximately -0.087 s. This is one preliminary case, not a population-level accuracy result.

3.4 Measures and Evaluation Metrics

Detector evaluation uses precision, recall, mAP50 and mAP50-95, consistent with the framework's validation outputs [11]. Precision concerns the correctness of positive detections and recall concerns recovery of labelled objects. Mean average precision summarises precision-recall behaviour using specified localisation thresholds. These metrics are not interchangeable with classification accuracy or the proportion of videos producing correct timing.

For clearance case k, signed error is e_k = predicted_k - observed_k. Negative values indicate underestimation. Mean absolute error (MAE) averages the absolute differences, while root mean squared error (RMSE) gives greater weight to larger errors:

$$MAE = \frac{1}{N} \sum_{k=1}^{N} |e_k| \quad (3.7)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^{N} {e_k}^2} \quad (3.8)$$

Software verification examines response fields, independent arithmetic recomputation, monotonic FIFO estimates, positive finite values, repeated processing and expected error handling. These checks isolate calculation and application behaviour. End-to-end latency, detection inference time and preview-rendering time are separate measures. Officer-task validation considers completion, interpretation and error recovery. The completed numerical and software checks are reported in Chapter 6; live and user evaluations are identified as future stages.

3.5 Data Analysis

Detector results are analysed by dataset and model version. Table 6.2 uses the recorded v2 training-notebook validation outputs [16], and Table 6.1 reports the later checkpoint metadata. The manuscript comparison [14] is checked against those records before attribution. A result from one dataset or checkpoint is not substituted for another, and differences between v2 and v3 are not interpreted as a controlled model-improvement experiment.

Calibration analysis summarises matched event counts, category coverage, queue-position coverage and the crossing-gap distribution. Effective discharge values are grouped by class and position; the median is the representative statistic used in the calibration method described in the research paper [14]. Observation-based values and prototype parameters for sparse groups have different evidential roles. The linked 283-event analysis set defines the descriptive empirical coverage used here.

The three preliminary clearance cases are analysed individually and through MAE and RMSE. Their clips overlap calibration material and the cases precede the current detector, so the results describe development-stage agreement. No population-level significance test is applied. The stationary-queue case is examined separately because its larger error provides information about the time origin and discharge state that an average alone would obscure.

3.6 Ethics, Privacy, Security and Responsible AI

Traffic footage may contain faces and number plates. Shared illustrations should minimise identifying detail, and operational access should be limited to the research or traffic-management purpose. Dataset use must respect source permissions and licences. This thesis includes researcher-supplied dataset and application captures as illustrations of the workflow. These captures are separate from the proposed officer and live-field evaluations, for which participant approval and organisational permission remain necessary.

The prototype uses password hashing and token-based authentication for selected operations. Security requirements also include controlled media access, safe upload naming, retention limits and protected database credentials. These requirements are relevant to future deployment because uploaded road footage and stored analysis histories can contain sensitive information. The present software checks do not constitute a penetration test or comprehensive access-control assessment.

AI assistance is disclosed in Appendix B. Research responsibility remains with the candidate, including checking numerical results, sources, model versions and interpretations. TrafficEye presents estimates for officer review. An ambulance label and a numerical duration are advisory outputs whose suitability depends on the

observed traffic and category accuracy; they do not independently authorise a signal change.

3.7 Reproducibility and Research Artefacts

Reproducibility requires more than the application source. The detector checkpoint, class map, exact dataset export, split manifest, training configuration, camera profile, ROI, reference line, calibration logs and lookup must be retained together. A hash identifies a file version but does not establish the validity of its contents or annotation process.

The current checkpoint is identified by the recorded SHA-256 prefix 5f9fef285679, while the earlier August checkpoint is identified by 1c3495784601 [16]. These identifiers distinguish the model attached to a result. Dataset version, ROI, reference line, lookup and processing settings are also necessary for reproduction, since replacing any one of them may change the represented queue or timing output.

The dependency record in Chapter 5 documents the local analysis environment separately from model training. A clean installation and dependency lock are appropriate release checks because model loading, video decoding and database access depend on their software versions. Appendix A lists the artefacts and metadata needed for reproducing the recorded demonstrations and for documenting future experiments.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The system requirements concern an officer-facing analysis workflow with live CCTV as its intended source. The demonstrated implementation accepts uploaded footage and recording metadata, validates the configured view, detects the supported classes, constructs an ordered queue and presents timing advice. Results and recording metadata support later review. The same analysis responsibilities apply to the intended camera workflow, with ongoing frame intake and queue updates required for live operation.

Table 4.1 maps functional requirements to the implemented demonstration. Reliability, interpretation and emergency-recognition performance are evaluated separately from feature availability. For example, an emergency advisory function can be implemented while its category accuracy still requires targeted validation.

Table 4.1: Functional requirements of the project

ID	Requirement	Reported implementation
FR1	Authenticate an officer for analysis	Login and bearer-token analysis request
FR2	Accept video and recording metadata	Multipart video upload, date and time
FR3	Check configured camera compatibility	ORB/RANSAC view matching and profile selection
FR4	Detect supported classes inside ROI	YOLOv8n and bottom-centre polygon filtering
FR5	Build ordered queue and intermediate quantities	Reference-line ordering and position groups
FR6	Recommend normal green duration	Clearance plus 3.12-second margin
FR7	Provide ambulance-related advisory	Emergency candidate queue and timing response
FR8	Store and review analysis records	MongoDB records, history and statistics
FR9	Provide an annotated preview	Background rendering and media response
FR10	Show file-based frame playback	WebSocket playback of uploaded material

Source: TrafficEye application design and implementation records [16].

Non-functional requirements include repeatable calculations, responsive presentation, understandable intermediate values, recoverable errors, controlled access and maintainability. The current evaluation covers selected deterministic and application properties. Continuous-feed latency, camera reconnection, media access and officer interpretation need separate measurement under the intended deployment conditions. Performance targets should be set before those trials and related to the available hardware and decision interval.

Live operation also requires information freshness. An estimate should be associated with the camera and observation time so that an officer can distinguish a current queue from an earlier state. The recorded-video demonstration preserves the input for review; a continuous feed would instead require a defined update policy and handling of arrivals after a queue observation. These are deployment requirements arising from the same single-stream timing model.

4.2 Architecture of the TrafficEye Solution

TrafficEye uses a React single-page application (SPA) connected to a FastAPI application programming interface (API). Uploaded video is read through OpenCV and analysed through Ultralytics YOLO. The queue-calculation module reads the lookup CSV and returns intermediate quantities with the timing estimates. MongoDB stores analysis and officer records; the frontend receives JavaScript Object Notation (JSON) responses and displays results.

Figure 4.1 shows the implemented demonstration architecture. A live camera adapter is shown separately as an intended extension of the video-input layer. The detector, camera profile and timing parameters supply the analysis engine; FastAPI coordinates the result, MongoDB stores its summary, and React presents it to the officer. The recommendation remains a human-reviewed output.

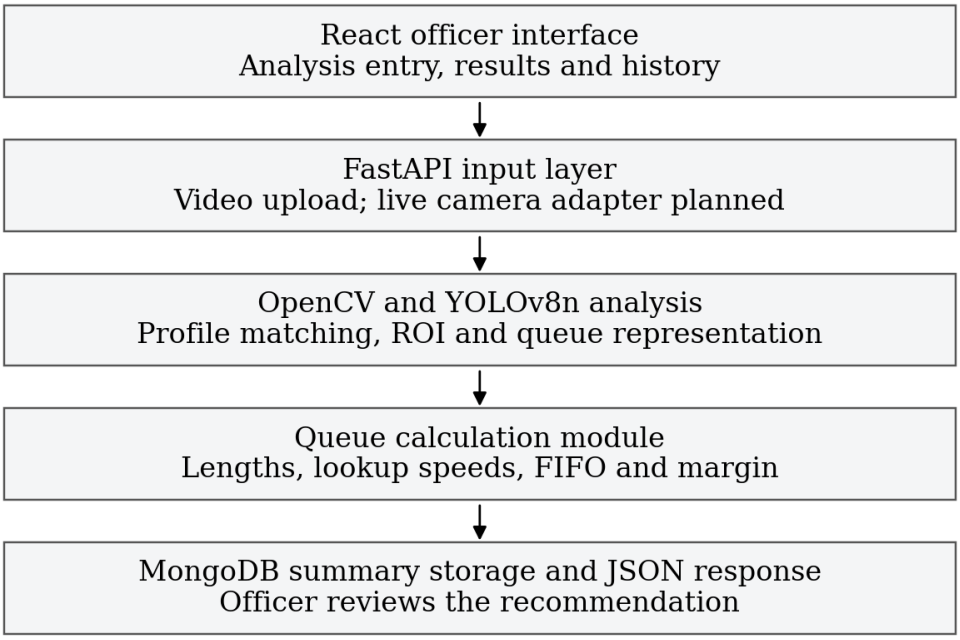


Figure 4.1: TrafficEye application architecture

Source: Author-created architecture diagram based on the TrafficEye implementation [16]. The live-camera adapter is a planned extension.

The architecture separates input acquisition from the queue calculation. Integrating a live CCTV source would provide ongoing frames to the detection layer, while the update policy would select the queue state to be analysed. The calculation module can then apply the same lengths, position groups, effective speeds and FIFO rule to that state. Camera interruption, stale results and scene changes must be handled explicitly during deployment. This is an internal extension of the independent TrafficEye project.

4.3 Design Alternatives and Decisions

The detector choice was informed by the recorded YOLOv5n-YOLOv8n comparison on v2 [16]. YOLOv8n produced higher precision, mAP50 and mAP50-95, while YOLOv5n had marginally higher recall in the reported validation outputs. The selection considers localisation performance and downstream queue representation. These results concern the v2 development experiment; the later YOLOv8n checkpoint remains a separate detector revision.

The timing design replaced a simple count-based rule with an explicit cumulative-distance and effective-speed calculation. The benefit of this design is that individual

vehicle contributions and lookup sources can be inspected. Its cost is reliance on calibration quality and several modelling assumptions. No measured superiority over the earlier rule is claimed without a matched clearance experiment.

Table 4.2 summarises the principal decisions. The recorded confidence threshold of 0.4 is an operating choice; a useful final evaluation would report precision and recall around this threshold rather than relying solely on aggregate validation mAP.

Table 4.2: Design decisions and their evaluation consequences

Decision	Selected approach	Consequence
Detector	YOLOv8n	Small model family; final-version evaluation required
Frame analysis	Every tenth frame	Lower sampling cost; may miss brief events
Queue snapshot	Highest ROI count	Simple selection; may not coincide with green onset
Queue order	Pixel distance to line_b	View-specific approximation, not metric geometry
Physical extent	Representative lengths plus common gap	Transparent estimate with parameter uncertainty
Discharge model	Class-position effective speeds	Needs support counts and provenance
Consistency	FIFO adjustment	Guarantees monotonic estimates, not accuracy
Camera acceptance	Feature matching and RANSAC	Blocks some unsupported views; thresholds need evaluation
User interaction	Officer advisory	Human review remains essential

4.4 Detailed Design

4.4.1 Processing stages

The demonstration processing stages are view validation, frame sampling, vehicle detection, ROI filtering, queue selection, spatial ordering, timing calculation and response creation. Normal and emergency candidates are handled separately. A valid duration requires both a usable represented queue and valid calculation parameters. An

incomplete queue or an unsupported speed entry should produce an explicit failure state, enabling the user to review the input or configuration.

Camera profiles describe the ROI polygon, reference line and reference image. Scene registration and normalisation support compatible views. This reduces the risk of applying a view-specific line and polygon to unrelated footage, but the method is not a substitute for a calibrated multi-camera deployment study.

4.4.2 Data representation

An ordered-queue entry includes vehicle type, queue position, position group, representative length, cumulative distance, effective speed, speed source, raw clearance and FIFO-adjusted clearance. The response also includes count summaries, queue length, normal clearance, suggested green and the safety buffer. Emergency-specific fields describe the associated advisory where available.

MongoDB stores recording metadata and summary outputs for history review. Vehicle-level ordered-queue detail is returned in the analysis response. Retaining that detail with the model, camera profile and lookup versions would strengthen retrospective reconstruction of a recommendation. The distinction between a summary record and a full calculation record informs the reproducibility requirements for a future release.

Table 4.3: Principal result fields and their meaning

Field	Meaning	Units or type
vehicle_count	Count used for the selected represented queue	Integer
vehicle_breakdown	Category counts	Mapping
queue_length_m	Estimated cumulative extent of last vehicle	m
clearance_time_s	Maximum FIFO-adjusted clearance estimate	s
suggested_green_s	Clearance plus safety buffer	s
ordered_queue	Intermediate vehicle-level calculation	Array
safety_buffer_s	Selected arithmetic margin	s
emergency_detected	Analysis emergency flag	Boolean
emergency timing fields	Timing estimate for emergency candidate queue	s

Field	Meaning	Units or type
annotated_video_url	Preview-media location	Path/URL

Source: TrafficEye API response contract [16].

4.4.3 Interface and feedback

The interface provides an operations overview, analysis results, an ordered queue, timing advice and history review. Queue clearance and the added margin are shown separately. Vehicle counts, category breakdown and intermediate values allow the user to inspect the calculation before interpreting the recommendation. The genuine interface captures are presented in Section 5.2.5 and Section 6.2.

These screens support the intended officer workflow, but the availability of an interface does not establish measured usability. A future officer-task study should examine interpretation of units, emergency status and unsupported inputs under realistic decision conditions. Live CCTV remains the intended source; the current screenshots illustrate the recorded-video demonstration.

4.5 Standards and Professional Practice

The design uses structured JSON, explicit units and identifiable error states. Vehicle classes and position groups are validated before timing computation. Intermediate values support numerical review. Observation-derived parameters and sparse-group prototype values remain distinguishable in calculation metadata.

Deployment requires route-level access checks, controlled media retention and safe uploaded-filename handling. Password hashing and token login are implemented, while complete access coverage and attack resistance require separate assessment. The prototype has not undergone comprehensive security certification.

Numerical results are attributed to their model and dataset, adopted parameters are referenced, and planned studies are identified. Diagrams explain the design; charts report the corresponding tables. This distinguishes detector performance, preliminary timing observations and intended deployment.

5 IMPLEMENTATION, INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

Model training and local application execution use separate environments. Development training used Google Colab GPU resources, including a Tesla T4 for the earlier comparison. The application environment uses Python with CPU-based PyTorch for local inference. The current checkpoint records a GPU training device and Ultralytics 8.4.173. Table 5.1 documents the selected local runtime versions, which are distinct from the training configuration.

The installed versions in Table 5.1 document the application environment. Training and application environments have different roles: GPU training produces the checkpoint, while the local runtime loads it for video analysis. Checkpoint compatibility, dependency locking and a clean-installation smoke test are therefore release concerns. A software-version record also helps identify whether a changed output follows a model revision or a processing-library change.

Table 5.1: Reported application environment

Layer	Technology	Reported version
Backend runtime	Python	3.12.8
API framework	FastAPI / Uvicorn	0.136.1 / 0.46.0
Detector runtime	Ultralytics / PyTorch	8.4.132 / 2.11.0 CPU
Video processing	OpenCV	4.13.0.92
Numerical analysis	NumPy / pandas	2.4.4 / 3.0.5
Database drivers	Motor / PyMongo	3.5.1 / 4.8.0
Validation models	Pydantic	2.13.4
Frontend	React / Vite	18.3.1 / 5.4.21
Frontend runtime	Node.js	20.15.1
Styling	Tailwind	4.3.3

Source: Local environment inventory [16]. Current checkpoint training metadata records Ultralytics 8.4.173.

The difference between a checkpoint's training-library version and the application's installed version should be retained in the final environment record. The code should

be tested with the chosen checkpoint rather than assuming that a later model file remains behaviourally identical to an earlier one. Dependency locking and a clean-environment smoke test are proposed for the final package.

5.2 Component Implementation

5.2.1 Video intake and camera profile

The main analysis endpoint accepts a video file, recording date and recording time, with optional location, road, direction, coordinates and notes. A bearer token is required for this analysis request. OpenCV reads the uploaded video, and the backend selects a compatible camera profile using reference-image features and RANSAC-based geometric agreement.

Unrecognised views are blocked with a calibration-required response rather than being processed using an arbitrary default ROI. Zero detections and failed scene registration can also lead to a blocked result. These behaviours support bounded use of a view-specific model, although their rejection accuracy has not been measured through a complete held-out view test.

5.2.2 Detector and selected queue

The operational detector samples every tenth frame at confidence 0.4 and filters bounding-box bottom-centres by the ROI. The backend records the sampled frame with the greatest in-region count as the normal queue candidate. It then sorts the detections relative to line_b and maps each position to its lookup group.

The detector performs sampled per-frame inference in the demonstration workflow. Offline ByteTrack processing links queue and crossing events for discharge calibration. The application's selected queue is a spatial snapshot rather than a continuously maintained track set. Future live-camera integration can extend queue-state handling with persistent identities and a defined update interval, but this behaviour is not reported as a completed field test.

5.2.3 Calculation and response

Queue estimates use the distance, speed and FIFO equations in Section 3.3. The response distinguishes the full-scene detection count from the vehicles represented inside the ROI. Representative queue length, clearance, margin and recommended green are returned with the ordered calculation. The recommendation is a timing calculation supported by detector observations; the interface label "AI Recommendation" does not mean that a language model generates the duration.

The backend creates the analysis record, stores summary information in MongoDB and returns the numerical response to React. Preview generation is handled separately and can remain pending after calculation completes. Raw class and confidence outputs provide the basis for detector validation; annotated media provide a visual inspection aid. This distinction is important when assessing emergency recognition, because displayed labels alone do not establish category accuracy.

5.2.4 Ambulance-related behaviour

The backend records an emergency candidate when an in-ROI Ambulance category is detected and calculates advice up to the frontmost ambulance in that candidate queue. Its correctness depends on class recognition and queue representation. An emergency status displayed in a screenshot is an application output, not independent confirmation of a real ambulance.

The ambulance-related calculation uses the FIFO-adjusted estimate at that vehicle's represented position and adds the same 3.12-second margin. Normal full-queue advice remains available. Validation should examine raw detections, missed ambulances and false emergency candidates together with the timing response. The supplied interface images document normal analysis and history; they do not provide an ambulance-specific field validation.

5.2.5 Interface implementation

The implemented React interface presents the analysis workflow through an overview dashboard, a detailed recommendation and a reviewable history. Figures 5.1-5.3 are researcher-supplied captures of that interface. They document visible output and

controls in the demonstration; the numerical values are not additional ground-truth traffic measurements.

Figure 5.1 shows summary cards for analysis records, detected vehicles, the latest queue count and emergency flags, together with recent analyses and a timing recommendation. The captured latest record shows 11 represented vehicles, a clearance estimate of 28.338 s and a recommendation of 31.46 s. These are application-record values rather than a count of independently verified traffic events.

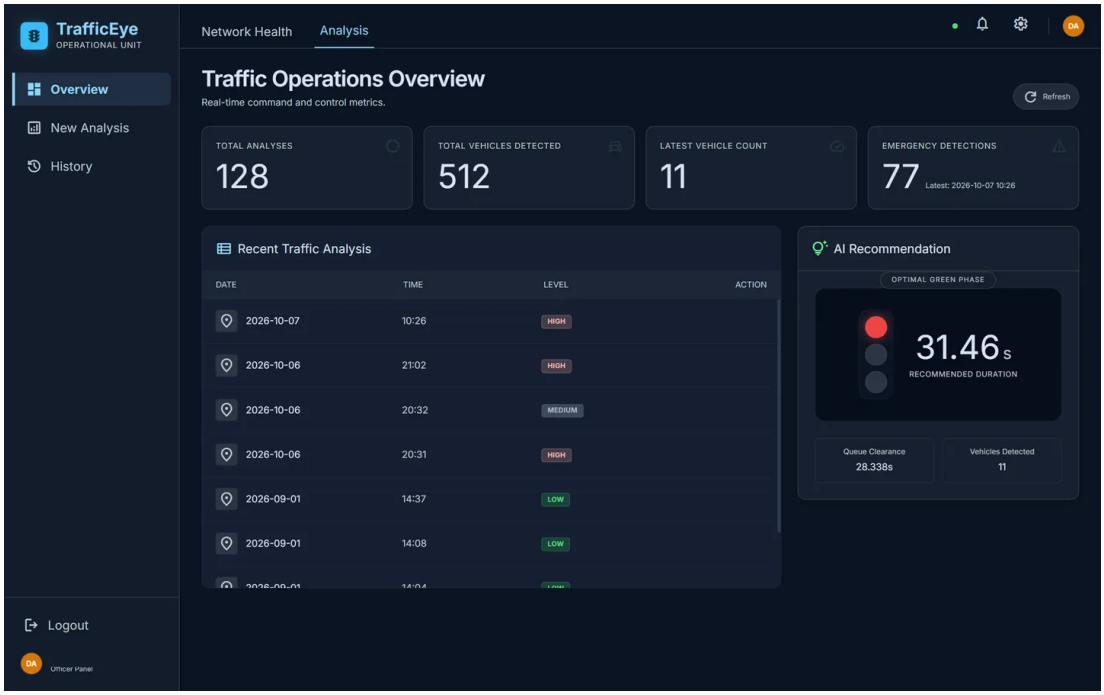


Figure 5.1: TrafficEye overview dashboard (original in colour)

Source: Researcher-supplied TrafficEye interface capture.

Figure 5.2 presents the annotated video beside the timing card. In this example, the clearance estimate is 23.5315 s for 10 represented ROI vehicles, with an added 3.12 s margin and a displayed recommendation of 26.65 s. The traffic-light icon visualises advice. The label "Optimal Green Phase" is interface wording; the study has not established a globally optimal junction policy or automatic signal actuation.

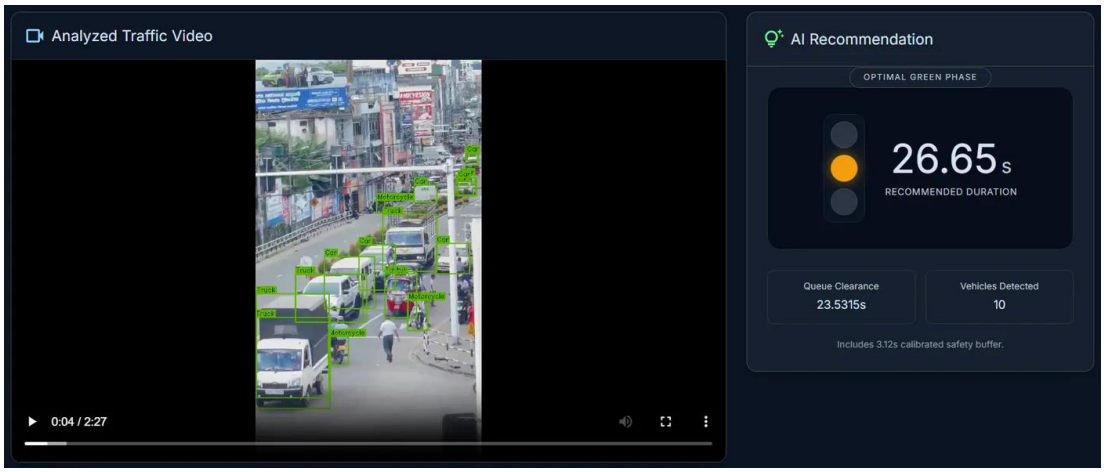


Figure 5.2: Annotated video and green-duration advisory (original in colour)

Source: Researcher-supplied TrafficEye recommendation capture.

Figure 5.3 shows the history screen, including date-range, traffic-level and emergency-status filters. Rows present stored count, queue-length estimate, clearance, suggested green, emergency flag and completion status, with a details link. Refresh and Export Log controls are visible; their presence in the capture is not treated as an independently completed functional test. History supports review of stored results, and an emergency flag retains the detector-based meaning described in Section 5.2.4.

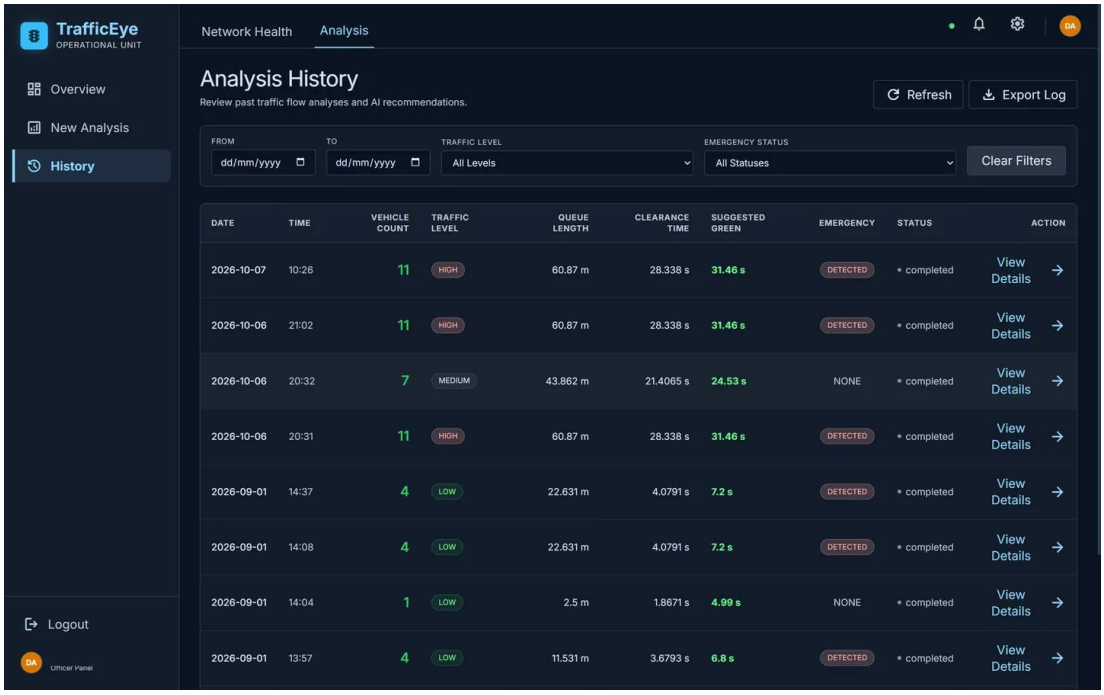


Figure 5.3: Analysis history and review controls (original in colour)

Source: Researcher-supplied TrafficEye history capture.

5.3 Internal Integration of Application Modules

Within TrafficEye, React, FastAPI, the queue-calculation module and MongoDB form the integrated application. The frontend submits recording metadata and video, the backend performs view-specific analysis, and the queue module returns intermediate and summary quantities. The database supports saved-result retrieval, while the frontend uses the returned fields to present normal and emergency advice.

Integration checks examine this data path rather than connections to unrelated research components. The response contract defines vehicle counts, estimated queue length, clearance time, green duration, margin, ordered-queue detail and emergency fields. Preview readiness is handled separately so that numerical results and media status can be represented accurately when they complete at different times.

For a future live camera adapter, the internal contract should include camera identity, observation timestamp, processing status and result freshness. Camera disconnection, unsupported scenes and calculation failure must produce distinguishable states. Continuous-feed integration extends the existing input and result contracts; it does not require the thesis to claim an already tested physical signal-controller connection.

5.4 Verification and Software Testing

Verification used recorded scripted acceptance and diagnostic checks [16]. They covered selected API requests, calculation recomputation, response invariants, repeated processing and synthetic queue inputs. These records describe development-stage software behaviour. Model-dependent checks should be repeated after a detector update, because consistent arithmetic does not ensure that a revised model detects the same vehicles.

The recorded API checks independently recomputed clearance and recommended-green values from returned ordered-queue entries. Response invariants checked units, finite outputs and consistent summary fields. A repeat analysis of clip_02 produced the same recorded result. Together, these checks support the implementation of the timing equations under the tested configuration, while detector-dependent accuracy requires separate evaluation.

Table 5.2: Recorded verification evidence and version boundary

Test	Recorded observation	Boundary
API acceptance	HTTP 200 for four recorded requests	Development requests; earlier detector
Arithmetic recomputation	Four of four responses matched	Existing lookup and equations
Response invariants	17 checks for each of four responses true	Recorded audit conditions
Repeatability	clip 02 repeated with identical output	Development audit; same input and configuration
Calculation stress	17 synthetic queues passed invariants	Calculation-only inputs
Frontend contract	17 expected fields present	Does not establish usability
Clearance observation	Three cases; MAE 2.511 s	Overlapping calibration clips
Emergency diagnostics	False candidates in eight normal clips	Earlier August detector

Source: Recorded acceptance and calculation-regression checks [16].

Synthetic inputs isolate the timing function from image perception. Longer lists of Bus, Truck and mixed vehicle categories check finite output and monotonic clearance behaviour. Such tests help identify arithmetic or lookup failures without requiring a matching video for every input combination. They are calculation tests and are interpreted separately from measured clearance intervals.

Further verification should exercise the chosen final model together with its camera profile and lookup. Non-functional evaluation should measure upload-to-result time, ongoing frame latency for live operation, repeated-run stability, upload limits, recovery from interruption and access to stored media. These are specified as future release and field checks rather than recorded passes.

Table 5.3: Error handling reported by the implementation

Condition	Reported response	Interpretation
Unreadable video	HTTP 400	Input cannot be analysed
Unauthorised main analysis request	HTTP 401	Authentication required
Unmatched camera view	HTTP 422 calibration_required	Profile or calibration needed

Condition	Reported response	Interpretation
Unsupported class / missing lookup	Blocked calculation, HTTP 422	No valid timing produced
No usable ROI vehicles after retry	HTTP 422	No supported queue result
Database-save failure	HTTP 500	Result persistence failed

Source: TrafficEye API and queue-module error contract [16].

5.5 Deployment and Operation

The demonstrated application runs locally with a FastAPI backend on port 8000 and a Vite frontend. MongoDB Atlas supplies database storage. This arrangement allows development and presentation without installing the software at a road intersection. Remote database storage is part of the data layer, while hosting the analysis service and connecting live cameras are separate deployment activities.

During demonstration, the user starts the configured application, signs in, uploads compatible footage, inspects the represented queue and recommendation, and reviews the saved summary. The recording supplies a repeatable input for showing the system's functions when live CCTV access is unavailable. A separate detector-comparison demonstration illustrates model outputs and is not an end-to-end performance benchmark for TrafficEye.

The demonstrated playback view streams frames from an uploaded file through a WebSocket. Live CCTV remains the intended field input. A future deployment would connect an authorised camera feed, verify its ROI and reference line, define ongoing queue updates and handle camera interruptions and stale estimates. These requirements relate the current demonstration to the intended real-time use without treating recorded playback as a completed live-camera trial.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 Current final-detector checkpoint

Table 6.1 presents the validation metrics recorded in the current YOLOv8n checkpoint saved on 6 October 2026 [16]. Precision is 0.7595, recall is 0.6788, mAP50 is 0.7477 and mAP50–95 is 0.5148. These are detector-level checkpoint metrics. The later export and training configuration are distinct from the development comparison reported in Section 6.1.2.

Table 6.1: Current October detector checkpoint metrics

Metric	Value	Evidence level
Precision	0.7595	Embedded checkpoint metric
Recall	0.6788	Embedded checkpoint metric
mAP50	0.7477	Embedded checkpoint metric
mAP50-95	0.5148	Embedded checkpoint metric

Source: Current YOLOv8n checkpoint metadata [16], saved 6 October 2026; SHA-256 prefix 5f9fef285679.

Precision and recall indicate the balance between correct detections and recovery of labelled vehicles, while mAP evaluates localisation across specified overlap thresholds. The values describe detector validation, not queue-clearance accuracy or live-road impact. Their interpretation depends on the associated labelled split, camera conditions and class distribution. They are therefore retained as a separate result set rather than merged with the development comparison.

6.1.2 Development detector comparison

Table 6.2 reports the recorded seven-class v2 validation outputs for YOLOv5n and YOLOv8n [16]. The split contains 599 validation images and 1,337 annotated instances. YOLOv8n reports precision 0.802, recall 0.794, mAP50 0.845 and mAP50-95 0.594; YOLOv5n reports 0.769, 0.796, 0.836 and 0.537 respectively. YOLOv8n has higher precision and both mAP measures, while YOLOv5n recall is higher by 0.002. The mAP50-95 difference is 0.057, or 5.7 percentage points.

Table 6.2: Recorded detector validation on the earlier v2 dataset

Model	Precision	Recall	mAP50	mAP50–95
YOLOv5n	0.769	0.796	0.836	0.537
YOLOv8n	0.802	0.794	0.845	0.594

Source: Recorded v2 training-notebook validation outputs [16]. Values are rounded to three decimal places; this is not the October v3 checkpoint.

The accompanying manuscript [14] lists a different YOLOv8n result row, which the project checkpoint records associate with an earlier six-class run. That row is not used as the seven-class v2 comparison in this thesis. Table 6.2 and Figure 6.1 use the v2 validation records consistently. The common dataset, epoch limit, image size and batch size support a development comparison, while framework-specific optimizer settings and shared recording sources limit broader claims of model superiority.

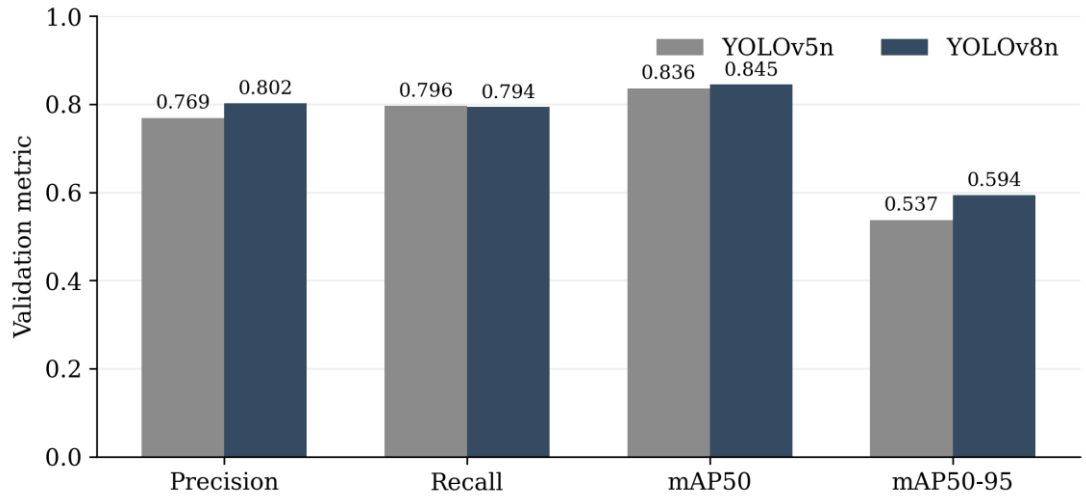


Figure 6.1: Recorded YOLOv5n and YOLOv8n v2 validation results

Source: Drawn from Table 6.2 and the v2 validation records [16].

6.1.3 Calibration coverage and safety margin

The linked calibration analysis contains 283 cleaned events spanning 15 of the 28 class-position groups. Events are concentrated at the front of the represented queue: 191 at P1, 74 at P2, 17 at position 3 and one at position 4. The P5+ group has no direct events in this linked analysis set. These counts describe observed calibration coverage; they do not imply that longer queues never appear in the recordings.

Crossing-gap analysis produced 289 valid intervals, a median of 2.90 s and an 85th percentile of 6.02 s. Their difference is 3.12 s, which lies within the project's 3-to-8-

second bounds. Table 6.3 records the derivation. The margin is a project-derived addition to clearance estimates, while its ability to compensate for errors under different traffic conditions requires further validation.

Table 6.3: Safety-margin derivation

Quantity	Value
Valid consecutive-crossing gaps	289
Median gap	2.90 s
85th percentile	6.02 s
90th percentile	6.517 s
P85 minus median	3.12 s
Project clamp bounds	3 to 8 s
Resulting margin	3.12 s

Source: Project crossing-gap analysis and margin summary [16].

6.1.4 Preliminary clearance observations

Table 6.4 presents the three recorded clearance cases from the earlier August workflow. Their recordings also contribute to calibration, so these are preliminary overlapping cases. They should not be described as held-out final-model validation.

Table 6.4: Recorded clearance estimates and observations

Case	Predicted (s)	Observed (s)	Error (s)
clip_01, frame 280	6.1601	13.464	-7.304
clip_02, frame 4090	4.4132	4.500	-0.087
clip_11, frame 3040	8.7887	8.932	-0.143

Source: Recorded preliminary clearance-evaluation outputs [16]. Errors are signed prediction minus observation.

The recorded summary gives MAE 2.511 s and RMSE 4.218 s. Two cases have small absolute errors, whereas the stationary-queue clip_01 case is underestimated by approximately 7.3 s. The notes associate this case with an observation taken before discharge began, illustrating the importance of the time-origin definition.

The three cases illustrate why aggregate error should be accompanied by individual observations. The larger stationary-queue error dominates RMSE, whereas two

moving-queue cases are close to their observed intervals. Because the recordings also contributed to calibration and predate the current detector, the results are treated as preliminary development observations. A held-out evaluation of the current complete workflow remains a separate research task.

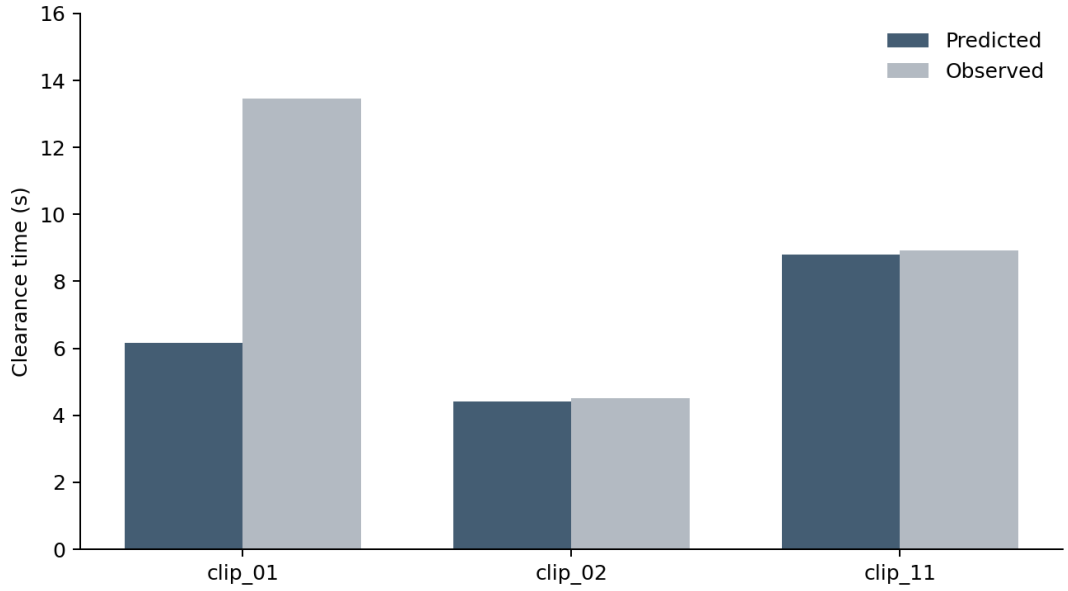


Figure 6.2: Three preliminary clearance cases

Source: Redrawn from Table 6.4. Calibration-overlapping cases, earlier August detector, N = 3.

6.2 Product and Technical Results

Development-stage application checks obtained successful API responses for four selected requests, matched four independent calculation recomputations, and satisfied the recorded response invariants. Repeating one clip after a different analysis produced the same recorded result. These observations support numerical repeatability and consistency of the application contract under the tested inputs [16].

The seventeen synthetic-queue cases produced finite positive values and preserved FIFO consistency. Table 6.5 provides selected examples. Their role is calculation regression: the values are generated from synthetic ordered categories, not measured road queues.

Table 6.5: Selected synthetic calculation-regression outputs

Synthetic queue	Clearance (s)	Interpretation
Car x 1	1.9963	Single-vehicle arithmetic input

Synthetic queue	Clearance (s)	Interpretation
Car x 5	8.7887	Five ordered Car inputs
Car x 10	17.9722	Longer synthetic queue
Car x 15	27.1556	Calculation stress input
Bus x 15	83.0903	Calculation stress input
Truck x 15	70.2674	Calculation stress input
Mixed queue, 10 vehicles	29.1327	Recorded mixed synthetic input

Source: Recorded synthetic queue-regression checks [16]. No video perception is involved in these inputs.

Figure 6.3 documents a displayed analysis result. The screen reports 18 full-scene detections and 10 vehicles represented in the ROI queue, a representative queue length of 53.693 m, clearance of 23.5315 s and recommended green of 26.65 s. The analysed-video preview, category breakdown and ordered-queue table connect the visible observations to the timing output. The preview is a playback view and should not be assumed to show the same frame used for the selected queue calculation.

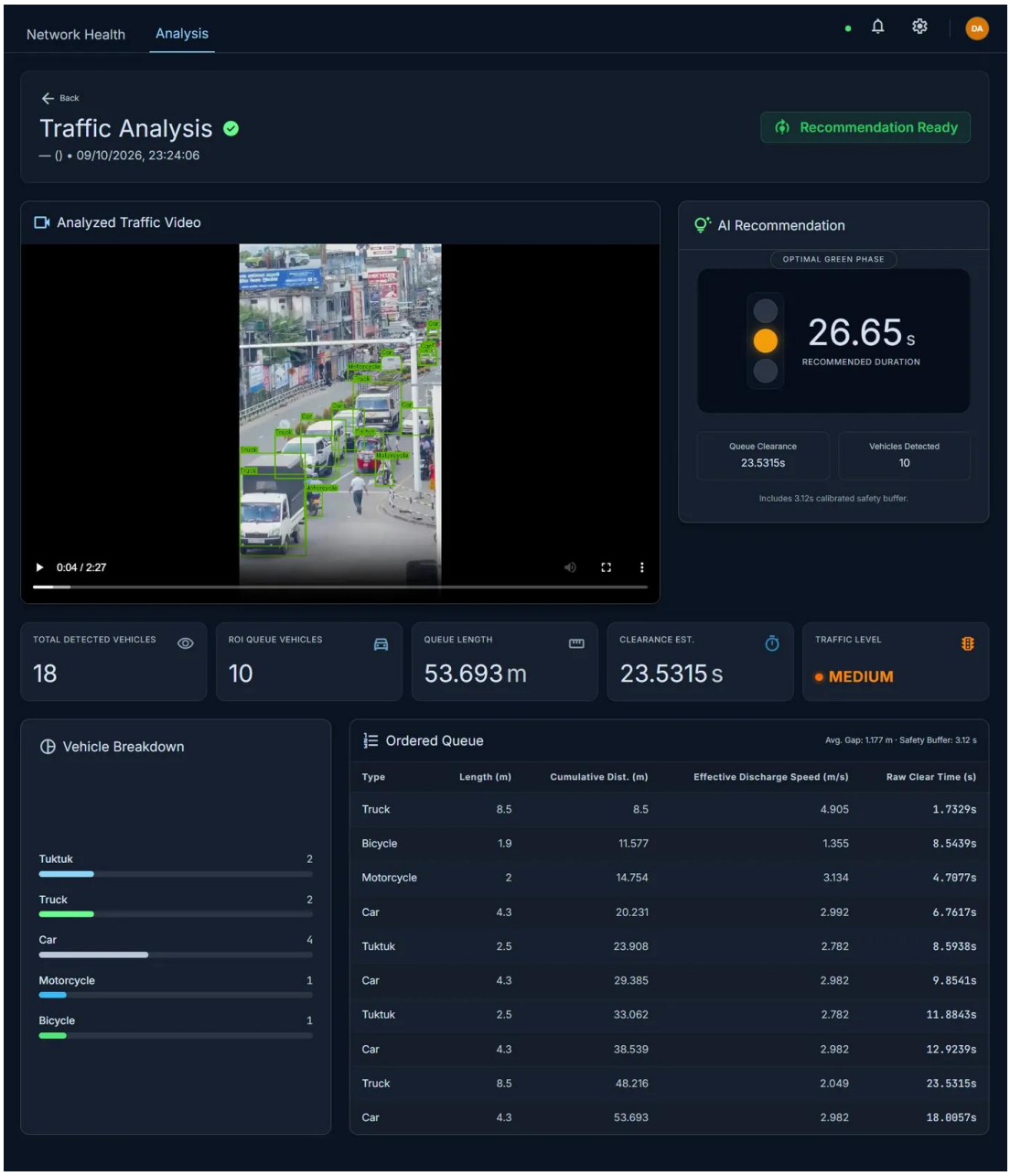


Figure 6.3: Detailed analysis output and ordered queue (original in colour)

Source: Researcher-supplied TrafficEye result capture; displayed values illustrate the application output.

Table 6.6 transcribes the ten ordered entries shown in Figure 6.3 so that the calculation can be read at normal document scale. Their representative lengths total 43.1 m; nine gaps of 1.177 m add 10.593 m, giving 53.693 m. The ninth vehicle has the largest displayed raw clearance, 23.5315 s. Although the final Car has a raw estimate of 18.0057 s, FIFO retains the preceding maximum for its adjusted clearance. Adding

3.12 s gives 26.6515 s, displayed as 26.65 s. This checks consistency of the displayed quantities; no observed clearance interval is supplied for this screenshot.

Table 6.6: Ordered queue transcribed from the displayed result

Pos.	Class	Cumulative distance (m)	Displayed speed (m/s)	Raw time (s)
1	Truck	8.500	4.905	1.7329
2	Bicycle	11.577	1.355	8.5439
3	Motorcycle	14.754	3.134	4.7077
4	Car	20.231	2.992	6.7617
5	Tuktuk	23.908	2.782	8.5938
6	Car	29.385	2.982	9.8541
7	Tuktuk	33.062	2.782	11.8843
8	Car	38.539	2.982	12.9239
9	Truck	48.216	2.049	23.5315
10	Car	53.693	2.982	18.0057

Source: Figure 6.3. Speeds and times retain the precision visible in the interface; raw times precede FIFO adjustment.

The available runtime observations concern individual development situations rather than a controlled product benchmark. Batch detector inference and frame playback measure different portions of the pipeline. No general real-time throughput or end-to-end latency value is assigned to TrafficEye in this thesis. Future camera trials should record hardware, input resolution, frame rate, analysis interval, computation time and result age together.

Emergency diagnostics during earlier development identified false Ambulance candidates in normal traffic footage. That observation motivates targeted validation of the current model with both real ambulances and visually similar ordinary vehicles. The present checkpoint metrics are aggregate detector measures and do not establish an ambulance-specific operating threshold or reliable emergency-advisory accuracy.

6.3 User or Stakeholder Validation

Structured officer-task validation has not yet been conducted in the reported study. The intended user role nevertheless informs the interface: a traffic officer must be able

to understand the represented queue, distinguish clearance time from the added margin, and identify the meaning of an emergency advisory. Technical demonstrations establish the availability of these functions; they do not provide measured usability or stakeholder acceptance.

The proposed instrument in Appendix A asks users to upload a supported clip, inspect the ordered queue, distinguish clearance from green duration, interpret an emergency advisory and recover from an unsupported-view response. Recording task completion, misunderstanding and feedback would connect interface functions to practical usefulness. Participant recruitment and permission must precede such a study.

6.4 Integration Results

Internal integration results concern the path from uploaded footage and metadata to backend analysis, calculation output and result presentation. Recorded API checks and the frontend field-contract review support selected interactions within this independent application. Database-backed history supports later summary review. The integration evaluated here is among TrafficEye's software modules.

Live CCTV integration and physical signal actuation are separate from these results. No field trace or signal-controller trial is reported. Future integration evaluation should record camera identity, frame timestamps, result age, connection recovery and the handling of failed analysis. The intended camera-based use can then be assessed under continuous operation rather than inferred from a recorded-video demonstration.

6.5 Summary of Findings

TrafficEye combines seven-class detection, a class-position queue calculation, a web application and a 3.12-second margin. Recorded v2 validation and current v3 metrics concern separate experiments. Two preliminary cases have close timing estimates; a stationary-queue case is substantially underestimated. Scripted checks support selected consistency properties, and screenshots document presentation and history review.

These findings support prototype feasibility. Independent final-workflow timing evaluation, broader calibration, officer tasks and live CCTV trials are needed to assess operational effectiveness.

7 DISCUSSION

7.1 Interpretation of Findings

The prototype demonstrates a way to organise vehicle detections into an inspectable timing calculation. A user can follow the representative length, cumulative distance, lookup speed and FIFO time for each represented vehicle. This is a useful property for explaining a recommendation and locating calculation errors. It does not establish that the model's assumptions accurately describe every observed queue.

The recorded v2 validation favours YOLOv8n on precision, mAP50 and mAP50-95, with marginally higher recall for YOLOv5n. These results concern one dataset split and framework-specific training settings. The current checkpoint uses a later export and configuration, so it is interpreted separately. The manuscript's earlier YOLOv8n row is not substituted for the recorded seven-class comparison.

The clearance cases are especially informative because their errors are not uniform. Two estimates are close to their observations, whereas one is substantially short. The stationary snapshot and the calibration time-origin definition provide a plausible explanation for the discrepancy, supported by the recorded notes. This suggests that alignment of observation time and clearance time is a more important issue than reporting only a favourable mean error.

7.2 Comparison with Literature, Baselines and Existing Solutions

The queue-estimation studies [3], [4], [6] focus on reconstructing traffic state, and the video-based study [7] demonstrates the use of computer vision for queue estimation. The queue-informed control study [5] instead examines how queue information can change signal operation. TrafficEye's output connects an ordered mixed-vehicle representation to an explanatory timing advisory. Comparisons must account for the observation source, target quantity and whether the final user is a person or a physical controller.

The recorded v2 detector comparison provides a quantitative development baseline [16]. Its precision, recall and localisation measures inform detector selection, but aggregate mAP does not identify every failure important to timing. Repeated seeds,

class-level errors and evaluation on independent camera sessions would strengthen a later benchmark.

The historical count-based timing rule has not been evaluated on the same cases as the final queue method. A future baseline experiment should preserve the same time origin, clips, detector version and observed endpoints. It should compare absolute errors as well as the pattern of underestimation. Without that experiment, the thesis can explain the richer modelling representation but cannot claim measured timing superiority.

7.3 Research Contribution

The applied research contribution is an explicit mapping from class-labelled, spatially ordered vehicle detections to a queue-clearance estimate. It combines category-specific representative lengths, queue-position speed groups, a monotonic FIFO adjustment and a documented percentile-based margin. The intermediate values make the calculation open to inspection rather than presenting an unexplained duration.

The project also provides preliminary evidence about the difficulties of connecting detection to clearance. Sparse class-position groups, a stationary-queue error and differences between operational and calibration observations identify limits that should guide subsequent study design. The findings are therefore a development and evaluation contribution, rather than proof of a generally optimal green-light policy.

The margin procedure is traceable to recorded crossing gaps, while the detector experiments provide a historical within-dataset selection comparison. Both contributions depend on preserving versioned artefacts and avoiding unsupported attribution of one version's results to another.

7.4 Product Contribution

The product contribution is an integrated web workflow for video upload, configured-view validation, seven-class detection, queue calculations and record review. Figures 5.1-5.3 and Figure 6.3 show how the overview, timing advisory, history and ordered-queue details are presented. The backend also provides a distinct ambulance-related

calculation. The intended deployment is live CCTV support for officers, with the current footage-based demonstration supporting development and inspection.

This workflow is more developed than a standalone detector notebook because it includes input handling, error responses, persistence and presentation. However, product usefulness is not established merely by the number of screens or returned fields. User-task observations are needed to determine whether officers correctly interpret uncertainty, distinguish the margin from clearance and recognise unsupported inputs.

Recorded-video demonstration is a practical development setting for this student research project. It allows repeatable presentation of the implemented functions while a live road installation is unavailable. Live CCTV operation remains the intended product direction. The next product stage concerns continuous acquisition, queue updates, interruption handling and field validation, rather than a change from an unrelated project purpose.

7.5 Limitations and Threats to Validity

7.5.1 Construct validity

The quantity estimated is the clearance time of a represented queue under project assumptions. It is not a direct measurement of saturation flow, total junction delay or complete safe signal timing. The displayed queue length is calculated from representative dimensions and gaps, rather than measured from calibrated image geometry. Interpreting those outputs as different traffic constructs would overstate the method.

The calibration interval begins from its recorded queue observation or first appearance, while the operational snapshot is selected by highest count. These definitions can disagree with the start of actual discharge. A start-up interval can therefore be omitted from a case's estimate even when the arithmetic is correct.

7.5.2 Internal validity and data support

Calibration coverage is uneven across vehicle classes and queue positions. The linked 283-event set covers 15 of 28 groups, with no direct Bicycle or P5+ events. Prototype parameters for sparsely represented combinations are not equivalent to well-supported group medians. This limits the strength of conclusions about long queues and rare categories. Greater coverage and category verification are needed before claiming reliable timing throughout the complete lookup space.

The preliminary clearance recordings overlap the calibration material. Their errors therefore do not provide an independent estimate of generalisation to unseen traffic. They also precede the current detector. A held-out design must freeze the model, camera profile and lookup before evaluating new recordings, preserving a consistent queue-observation and crossing-time definition. Nominal ambulance-labelled calibration events require visual confirmation.

7.5.3 External validity

The monitored-view setup is narrow and camera-dependent. ROI shape, reference line, scene registration and pixel-distance order may not transfer to another road. Additional Sri Lankan-style training photos do not, on their own, establish generalisation across locations, weather, lighting or camera heights.

The development dataset includes related material from the same source videos across partitions. Related frames can make a validation result more favourable than evaluation on a new recording session. Dataset expansion with local images improves the range of examples but does not establish source-independent performance by itself. Future evaluations should report source-separated camera or session splits and class-specific detection outcomes.

7.5.4 Statistical and operational validity

With three preliminary cases, the summary error is strongly affected by one larger underestimate. The sample is insufficient for a stable population-level accuracy estimate or a statistical significance claim. Case-level errors are therefore reported

alongside MAE and RMSE. Similarly, isolated inference or playback measurements cannot establish complete-system latency under continuous camera operation.

FIFO simplifies traffic to one ordered stream, and the adopted gap is applied uniformly between representative vehicles. Motorcycle filtering and lane changes can violate that representation. The recommendation also does not calculate a complete signal plan with pedestrian, amber and all-red timing. Officer validation, live camera trials and traffic-outcome evaluation remain necessary before interpreting the advisory as evidence of reduced congestion, improved safety or faster emergency passage.

7.6 Implications

Further research should align calibration and evaluation time origins, preserve independent recording groups and increase support for sparse categories and positions. Individual-case errors and provenance should accompany aggregate metrics.

The detector, camera profile and lookup should form one reproducible release. Detailed result records can support review of changes in category recognition or timing. Raw detections should remain available alongside annotated previews. Officer evaluation and traffic-engineering review can then establish whether the transparent calculation is understandable and useful in the intended decision setting.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

TrafficEye's proposed value is to provide traffic officers with an inspectable queue-clearance estimate and green-duration advisory from a configured camera view. Potential users include traffic-management personnel and road authorities evaluating camera-based support. The intended service is live CCTV monitoring; recorded-video review also supports development, training and retrospective inspection. These use cases form a proposed adoption direction rather than a verified customer base.

The web interface makes the vehicle composition and calculation available for review instead of displaying an unexplained duration. During a pilot, that transparency can be assessed through task completion, understanding and the officer's ability to identify an unsuitable estimate. The current demonstration shows how the system works; the value of the information in operational decisions needs user and field evaluation.

8.2 Market and Competitive Position

The system is positioned as a research-stage traffic advisory tool. Its proposed role complements the officer's assessment of traffic rather than replacing junction-wide engineering or a certified controller. Comparison with alternatives should examine deployment effort, category support, explanation, maintenance and accuracy under the same site conditions. No market share or competitive ranking is inferred from the development experiment.

Potential differentiators to evaluate are explanation of intermediate values, support for the project's seven vehicle categories and camera-specific calibration. Barriers include detector reliability, maintenance of per-view calibration, data governance and integration with existing workflows. These points are design implications, not verified competitive advantages.

8.3 Business or Sustainability Model

A possible institutional adoption model combines an evaluated pilot, camera configuration, detector and lookup maintenance, and technical support. The prototype

was developed without an operational commercial contract. Pricing and return on investment are not estimated without a cost ledger and measured user benefit. Camera access, analysis hardware, annotation, storage and ongoing validation are the relevant resources for a future deployment.

Table 8.1 identifies cost categories that should be measured during a pilot. Sustainability requires accounting for staff time and ongoing evidence maintenance as well as computation. A system that needs repeated manual recalibration should make that cost visible.

Table 8.1: Proposed cost and maintenance categories

Category	Resource to measure	Pilot consideration
Data preparation	Images, annotation and version records	Record sources and model export
Computation	Training GPU and analysis machine	Measure hardware use and latency
Camera integration	Feed access, ROI and reference line	Evaluate each monitored view
Validation	Held-out clips and officer tasks	Measure accuracy and interpretation
Storage	Footage and analysis records	Set access and retention policy
Maintenance	Dependencies, model and lookup updates	Recheck complete workflow after changes

8.4 Scalability, Intellectual Property and Compliance

Scaling to more cameras would require profiles, representative evaluation material and an assessment of lookup transfer. Queue-ordering and discharge assumptions cannot be assumed to apply unchanged at every site. A controlled site-onboarding process would test view matching, class recognition and clearance intervals before enabling recommendations.

Commercial or institutional distribution requires review of the licences attached to the chosen software and dataset exports. Third-party packages, labelled images and pretrained weights may have different reuse conditions. This section identifies licence review as an adoption requirement, without providing a legal determination about an unexamined distribution arrangement.

Public-road video handling should define the permitted purpose, authorised users, retention period and removal of identifying information from shared examples. Organisational camera permission and the relevant institutional procedures should precede participant work and deployment. These are practical governance requirements for moving from a local demonstration to a live traffic-management setting.

8.5 Contribution to the Sustainable Development Goals

The intended research direction aligns conceptually with Sustainable Development Goal 11, particularly its transport-related target, and Goal 9's emphasis on infrastructure and innovation [17, 18]. A traffic-monitoring advisory could contribute to those directions if it is shown to improve a relevant workflow or transport outcome.

This prototype has not measured an SDG indicator, emissions reduction, safety improvement or access benefit. The connection is therefore a statement of potential relevance, not demonstrated impact. Future field or simulation evaluation should define a concrete outcome before sustainability benefits are quantified.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

TrafficEye was developed as an independent camera-based traffic-monitoring and green-light-support system. The intended operational source is a live CCTV feed from a configured road approach. Because a live installation was unavailable, prerecorded videos demonstrated vehicle recognition, ordered queue representation, clearance estimation and presentation of advice. The web application integrates these functions with recording metadata and result storage.

The method combines seven-class YOLOv8n detection, representative vehicle lengths, a published 1.177 m standstill gap, class-position effective discharge values and FIFO consistency. Crossing-gap analysis yields an additional 3.12-second margin. Recorded v2 validation supports the detector-development comparison; the later v3 checkpoint reports its own metrics. Preliminary clearance cases yield MAE 2.511 s and RMSE 4.218 s. Software records and the supplied result capture support selected numerical-consistency properties, while the screens document presentation and history review.

The main conclusion is that the implemented prototype provides an inspectable connection between vehicle observations and timing advice. The evaluation supports feasibility within the demonstrated setting. The preliminary timing sample, uneven calibration coverage and model-version differences limit conclusions about independent timing accuracy. Officer validation and live CCTV testing are the next stages for assessing the system's operational usefulness.

9.2 Achievement of Objectives

Table 9.1 relates each objective to the implemented functions and reported evaluation. Development achievement and strength of validation are distinguished so that completion of a feature is not treated as proof of general field performance.

Table 9.1: Objective achievement within the delivered scope

Objective	Outcome	Evaluation boundary
O1 Detector data and model	Roboflow v3: 3,370 images; seven-class YOLOv8n checkpoint	Development comparison and separate current metrics
O2 Ordered queue	ROI filtering and reference-line spatial order	Configured single-stream snapshot
O3 Clearance and margin	Cumulative distance, class-position speeds, FIFO and 3.12 s margin	Uneven calibration; preliminary timing cases
O4 Normal and emergency advice	Full-queue and ambulance-position recommendations	Targeted current-model emergency testing needed
O5 Web workflow	FastAPI, React and MongoDB application; supplied interface captures	Recorded-video demonstration completed
O6 Evaluation	Detector metrics, calculation regression and preliminary cases	Live CCTV and officer-task evaluation remain future stages

9.3 Individual Research and Product Contributions

My individual research contribution is the class- and position-aware vehicle-to-clearance method and its development evaluation. My product contribution is the integrated TrafficEye web application, including video intake, configured-view analysis, transparent timing outputs, ambulance-related advice and record review. The project was undertaken independently.

The thesis connects the problem, literature, adopted parameters, implementation and results into one account of the work. AI-assisted development and drafting are disclosed in Appendix B. The research contribution remains the implemented and explainable workflow, its supporting observations and the interpretation of its limitations, rather than the use of an automated writing or coding tool.

9.4 Future Directions

The immediate deployment direction is authorised live CCTV integration. This requires a defined queue-update interval, handling of new arrivals, camera interruption recovery and detection of stale results. The selected model and camera profile should then be evaluated together on continuous footage, including occlusion, ordinary vehicles resembling ambulances and verified ambulance examples.

Timing research should increase class-position coverage and use held-out recordings with a common observation-time definition. Sensitivity analysis can quantify the influence of representative lengths, the adopted standstill gap and the margin. A count-based baseline should use the same detected queues and observed endpoints, allowing the contribution of class and position information to be measured directly.

Further development includes officer-task validation, retention of full intermediate result records, stronger access and upload controls, and reproducible installation. Multi-lane queue handling would require a model of lateral movement rather than simply reusing single-stream FIFO. Traffic simulation or an authorised controller pilot could later examine delay, residual queues and emergency passage, but those outcomes are outside the completed demonstration.

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GLOSSARY

Specialised terms

Term	Meaning in this thesis
Calibration event	Temporally linked queue observation and reference-line crossing used to estimate an effective speed.
Camera profile	Configured reference view, ROI and line information used to accept and interpret footage.
Clearance estimate	Modelled interval associated with the last represented vehicle, not a measured complete signal cycle.
Effective discharge speed	Estimated cumulative distance divided by the adopted observed interval; not instantaneous measured road speed.
Emergency candidate frame	Sampled frame used for ambulance-related queue calculation; it may differ from the normal frame.
Fallback value	A prototype parameter used where an empirical group estimate is unavailable; it must retain its source label.
Held-out recording	A recording excluded from model training and parameter calibration before evaluation.
Queue snapshot	Selected frame and its ordered in-ROI detections.
Source-separated split	An image split that keeps related recording sources on one side of an evaluation boundary.
Validation metric	Performance summary on a specified labelled split; not proof of application or field effectiveness.

APPENDIX A RESEARCH INSTRUMENTS AND EXTENDED RESULTS

A.1 Extended Geometry Specification

Table A.1 documents how each representative length enters the timing model. Five category values are supported by the published Sri Lankan measurements [8], while Bicycle and Ambulance are prototype values. The adopted gap is taken from a separate published calibration study [2]. Geometry parameters are independent of image-space ordering, which identifies position without directly measuring road distance.

Table A.1: Geometry parameters and their sources

Parameter	Value	Source
Motorcycle length	2.0 m	Jayasinghe et al. [8]
Tuktuk length	2.5 m	Jayasinghe et al. [8]
Car length	4.3 m	Jayasinghe et al. [8]
Bus length	10.8 m	Large bus in Jayasinghe et al. [8]
Truck length	8.5 m	Heavy goods in Jayasinghe et al. [8]
Bicycle length	1.9 m	Jayasinghe et al. [8]
Ambulance length	5.5 m	Jayasinghe et al. [8]
Standstill gap	1.177 m	Gunarathne et al. [2], Table 4, trial 43

Source: Jayasinghe et al. [8], Gunarathne et al. [2] and TrafficEye prototype specification [16], [14].

A.2 Linked Calibration Distribution

Tables A.2 and A.3 report the 283 cleaned linked observations by category and position. They describe the temporal support available for the calibration analysis. Position counts refer to individual linked events; they do not describe the maximum queue length present in every video. The distribution informs the limitations discussed in Section 7.5.

Table A.2: Cleaned linked calibration events by class

Class	Events	Qualification
Car	111	Largest supported class group

Class	Events	Qualification
Motorcycle	72	Requires attention to mixed-traffic order
Bus	43	Limited position coverage
Tuktuk	28	Limited position coverage
Truck	27	Limited position coverage
Ambulance	7	Limited position coverage
Bicycle	5	Limited position coverage
Total	283	Cleaned linked analysis events

Source: Linked calibration-event records [16].

Table A.3: Queue positions of cleaned linked calibration events

Position	Events
P1	191
P2	74
Position 3	17
Position 4	12
P5+	7
Total	283

Source: Linked queue-position records [16].

A.3 Preliminary Clearance Evaluation Specification

Table A.4 records the preliminary evaluation boundary. The three development cases compare an estimated clearance interval with its recorded observation. Because calibration and case footage overlap, the aggregate error is descriptive rather than a held-out final-system result. Subsequent trials should freeze the model and lookup, identify the observed start and crossing endpoint, and retain the ordered queue for each estimate.

Table A.4: Preliminary evaluation conditions

Property	Recorded value	Interpretation
Case count	3	Preliminary development evaluation
Model period	Earlier August workflow	Not current-checkpoint evaluation

Property	Recorded value	Interpretation
Recording independence	Calibration overlap	Not a held-out result
MAE	2.511 s	Descriptive absolute error
RMSE	4.218 s	Sensitive to the larger case error
Error direction	All three negative	Underestimation in these cases

Source: Development clearance records [16].

A.4 API and Failure-Response Specification

The main endpoint is POST /api/analysis. Its multipart input includes file, recorded_date and recorded_time. Optional fields include location, road, direction, gps_lat, gps_lng and notes. The response contains the stored summary fields and the detailed ordered_queue array. Emergency timing and annotated-video status are included where applicable.

Supporting routes provide login, registration, current-user information, analysis history, record retrieval, statistics, preview readiness and file-playback streaming. Table A.5 specifies further verification scenarios and the outputs to retain. These scenarios are a test specification, not an assertion that every final-version case has already passed.

Table A.5: Proposed final-version verification cases

ID	Scenario	Expected evidence
PV1	Supported normal video	Raw response, ordered queue, version and preview
PV2	Real ambulance video	Raw detections and advisory calculation
PV3	Normal clips with ambulance-like objects	False-emergency count and frame review
PV4	Uncalibrated view	Recorded calibration_required rejection
PV5	Unreadable or invalid upload	Recorded input error without valid advisory
PV6	Repeated normal analysis	Outputs compared for repeatability
PV7	Missing or invalid lookup entry	Explicit blocked calculation

ID	Scenario	Expected evidence
PV8	Unauthorised requests and media access	Access-control outcomes for each route
PV9	Longer supported input	End-to-end latency and resource record
PV10	History review	Stored summary and recoverable intermediate details

Source: Proposed test design, not completed results.

A.5 Proposed User and Stakeholder Instrument

This instrument is proposed for an approved user-validation session. Participants should be told that the product is a research prototype and that its output is an estimate. No participant number, demographic profile or response is asserted in this draft.

The researcher should record the participant's relevant role, session conditions and consent status, avoiding unnecessary personal identifiers. The facilitator should provide the same task wording and keep intervention separate from independent task completion. Technical failures should be recorded rather than replaced with a successful demonstration.

Table A.6: Proposed user-validation tasks

Task	Action	Observation to record
U1	Log in and submit a supported clip	Completion, errors and assistance
U2	Explain the represented vehicle queue	Correct interpretation of order and class
U3	Distinguish clearance, margin and green duration	Misunderstanding or correct explanation
U4	Interpret an ambulance advisory	Understanding of advisory and uncertainty
U5	Handle an unsupported camera view	Recovery action and clarity of error
U6	Find a previous analysis	Completion and information availability

After the tasks, ask which information was clear, which information was confusing, whether the participant could explain the basis of the recommendation, and what additional information would be needed before practical use. Record comments verbatim where permission allows and separate them from researcher interpretation.

If a standard questionnaire such as SUS is used, its original wording, scoring method and source should be included in the final instrument. This draft does not reproduce a

questionnaire or report a SUS score without a completed, documented study. Suggested timing measures should be recorded as observations, not inferred from a participant's opinion.

A.6 Reproduction Package and Final Artefact Inventory

A reproduction package links each reported result to the dataset, detector, lookup and software configuration that produced it. Table A.7 specifies the artefacts and metadata needed for that relationship. Keeping the input recording and full response allows a result to be inspected independently of a later interface change or model update.

Table A.7: Expected reproducibility artefacts

Artefact	Purpose	Metadata to retain
Detector checkpoint	Reproduce inference	Model version, class order, file identifier
Dataset export	Reproduce training/evaluation	Sources, splits, annotation and export settings
Training record	Explain model configuration	Initial weights, resolution, batch size and optimizer
Camera profile	Reproduce ROI and queue order	Reference image, polygon and reference line
Calibration events	Reproduce discharge analysis	Clip, track, position and crossing timestamps
Timing lookup	Reproduce duration	Class, position group, speed and parameter source
Analysis response	Reconstruct a recommendation	Ordered queue, intermediate values and summary
Software environment	Reproduce application	Dependency versions and run configuration
Future field/user records	Evaluate operational use	Tasks, conditions, consent and observed outcomes

APPENDIX B ACADEMIC INTEGRITY AND AI USE DISCLOSURE

B.1 Scope of Assistance

AI tools assisted system development and thesis preparation. Claude Code was used for coding support, debugging and organisation of project records; ChatGPT and Codex assisted explanations, thesis structure, drafting and document preparation. Google Stitch assisted interface-design work. These tools supported the research workflow and do not replace the candidate's responsibility for understanding and verifying adopted outputs.

Assistance included reviewing code descriptions, organising model and calibration information, explaining calculations and revising academic prose. Generated suggestions were treated as development or drafting inputs, not as empirical observations. Table B.1 identifies the tool purposes and the checks required when using their outputs. No new detector experiment or field study is claimed on the basis of document preparation.

Table B.1: AI and automated assistance identified in the draft record

Tool	Purpose	Verification responsibility
Claude Code	Coding, debugging and project-record organisation	Review adopted code and raw numerical outputs
ChatGPT and Codex	Explanations, thesis structure and document drafting	Review scope, citations, numbers and conclusions
Google Stitch	Interface-design assistance	Review adopted design and implementation
Document and plotting tools	Layout and numerical graphics	Check tables, charts and rendered pages

B.2 Representative Workflow and Verification

Representative workflows included asking for explanations of the queue-clearance equations, inspecting project artefacts for model and dataset versions, and mapping the research to the individual-thesis template. Thesis revisions checked the distinction between intended live CCTV use and recorded-video demonstration, the independent-project scope, and the sources of adopted geometry parameters.

Conceptual diagrams explain the workflow, and numerical charts are drawn from the corresponding tables. Roboflow and application screenshots are supplied project captures, not generated substitutes for the interface. The v2 detector comparison is attached to the recorded validation outputs; current metrics are attached to the checkpoint; preliminary clearance results retain their development-stage boundary. The screenshot calculation is reported as displayed arithmetic, not a new ground-truth accuracy test.

Verification involves checking citations against the research sources, numerical tables against their associated records, and the document's headings, tables and pagination after rendering. Before signing the declaration, I am responsible for reviewing the final thesis and retaining representative AI prompts or workflows where required by the module policy.

B.3 Responsibility for Final Submission

I remain responsible for the detector configuration, dataset preparation, queue-event definition, calibration parameters, FIFO rule, margin and conclusions presented in this thesis. AI assistance does not establish that an estimate is accurate or that a future feature has been deployed. The results and limitations must be explained using the research methods and records.

Future model updates, new calibration data, live camera trials and officer studies should be documented as additional work with their own settings and results. The thesis distinguishes the completed prototype, development observations and intended deployment so that later revisions can be assessed against the appropriate evidence.